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(54) **MIGRATION OF WORKLOADS ACROSS
CLOUD SERVICES BASED UPON
ENDPOINT PERFORMANCE**

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(71) Applicant: **Dell Products, L.P.**, Round Rock, TX
(US)

(72) Inventors: **Anantha K. Boyapalle**, Cedar Park,
TX (US); **Vivek Viswanathan Iyer**,
Austin, TX (US)

(73) Assignee: **Dell Products, L.P.**, Round Rock, TX
(US)

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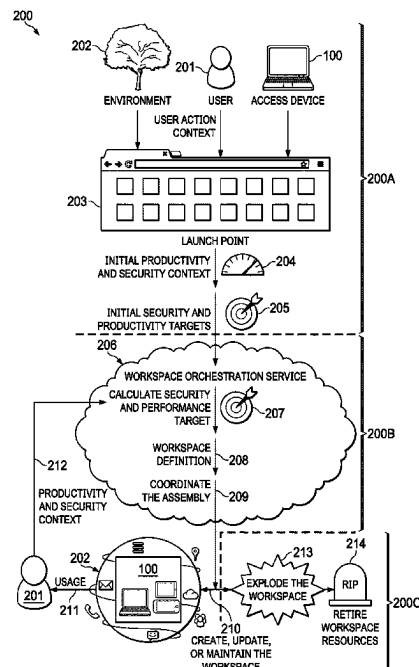
Primary Examiner — Sisley N Kim

(74) *Attorney, Agent, or Firm* — Fogarty LLP

(57) **ABSTRACT**

Systems and methods for migration of workloads across cloud services based upon endpoint performance are described. In some embodiments, an endpoint Information Handling System (IHS) may include a processor and a memory coupled to the processor, the memory having program instructions stored thereon that, upon execution, cause the IHS to: receive telemetry data indicative of a utilization of the IHS during execution of a workload by a first cloud service; determine a difference between a capability of the IHS and the utilization; and at least one of: (a) in response to the difference being greater than a first threshold, migrate the workload to a second cloud-based service with higher performance than the first cloud service; or (b) in response to the difference being smaller than a second threshold, migrate the workload to a third cloud-based service with lower performance than the first cloud service.

20 Claims, 6 Drawing Sheets



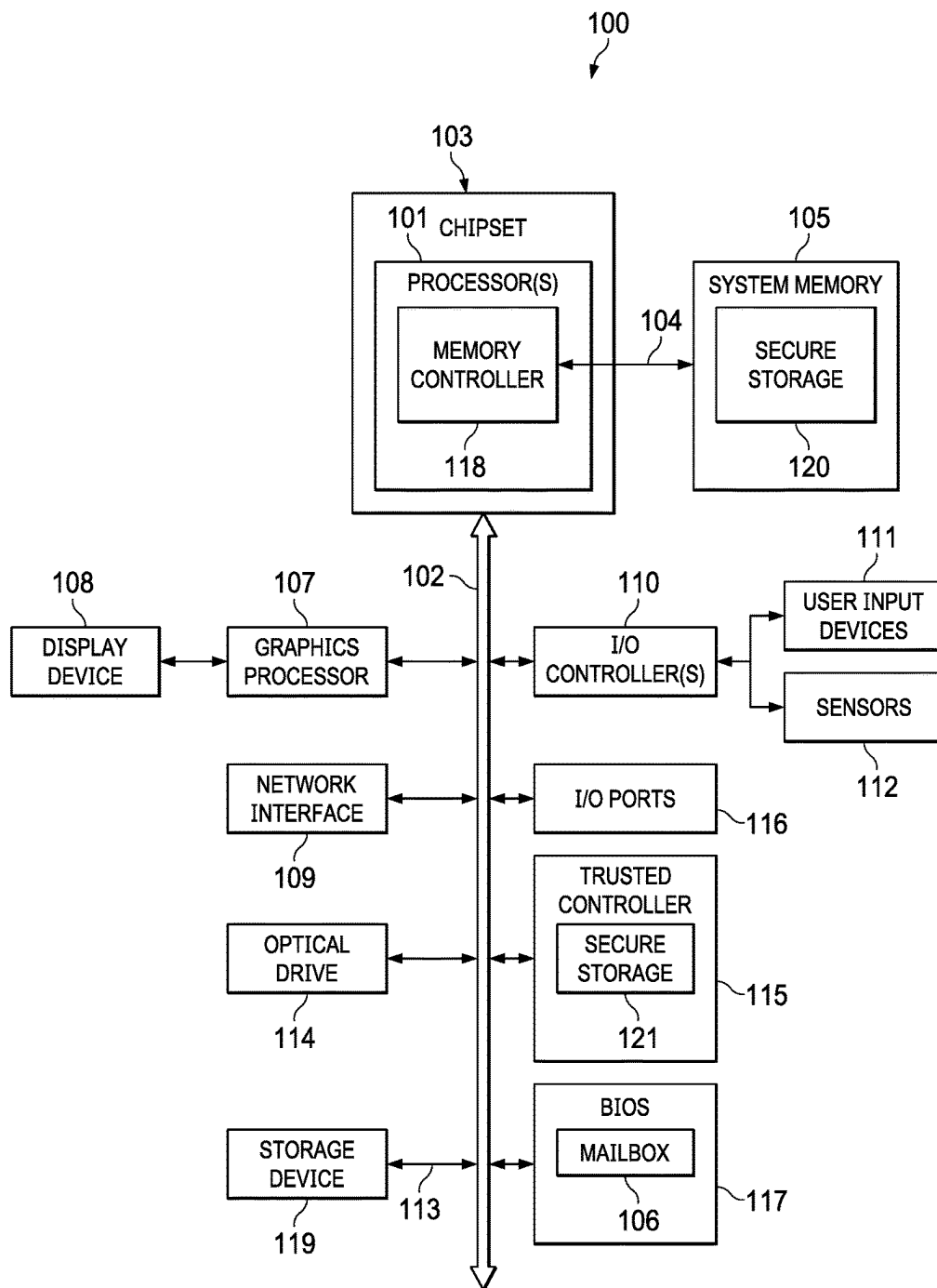
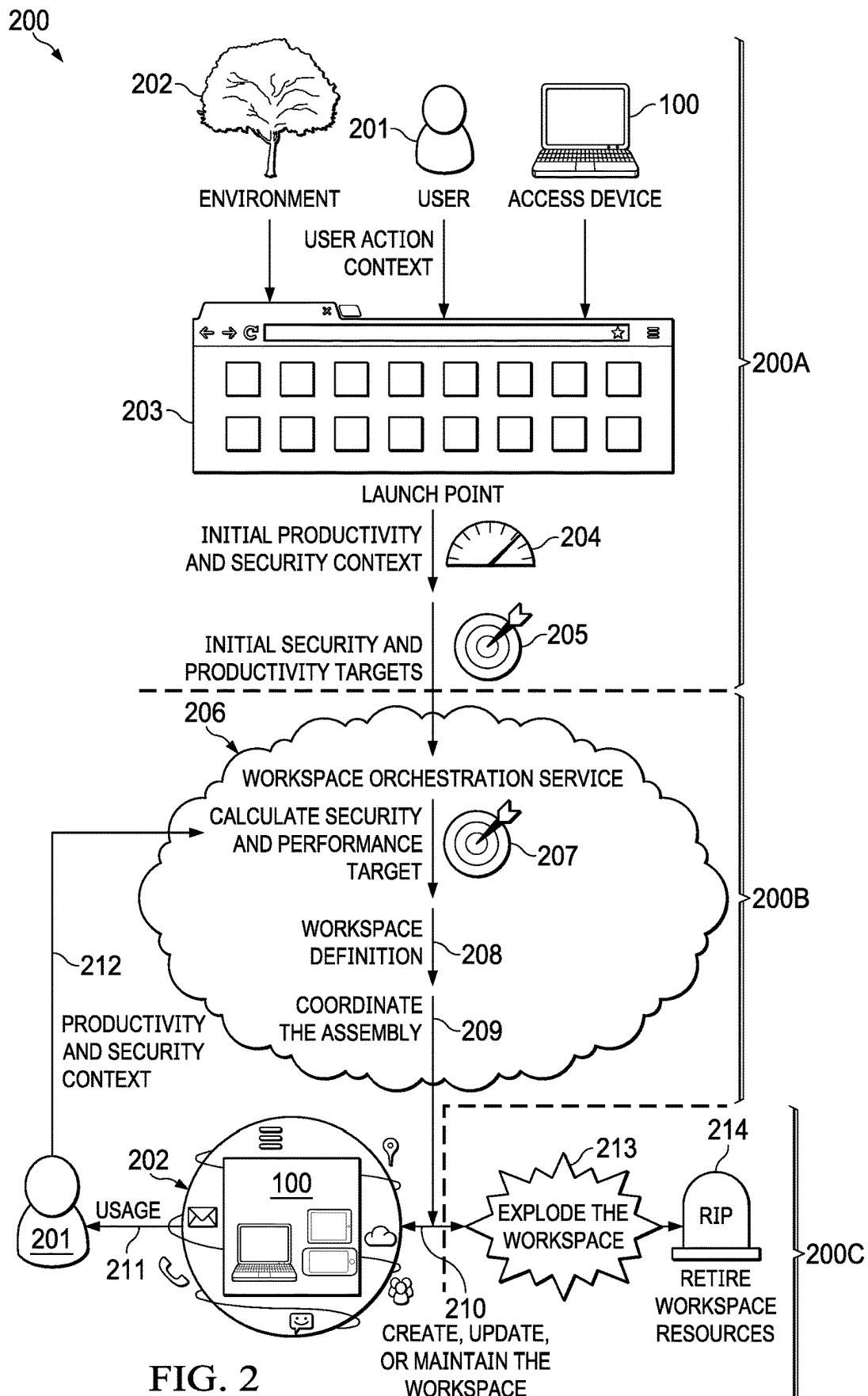
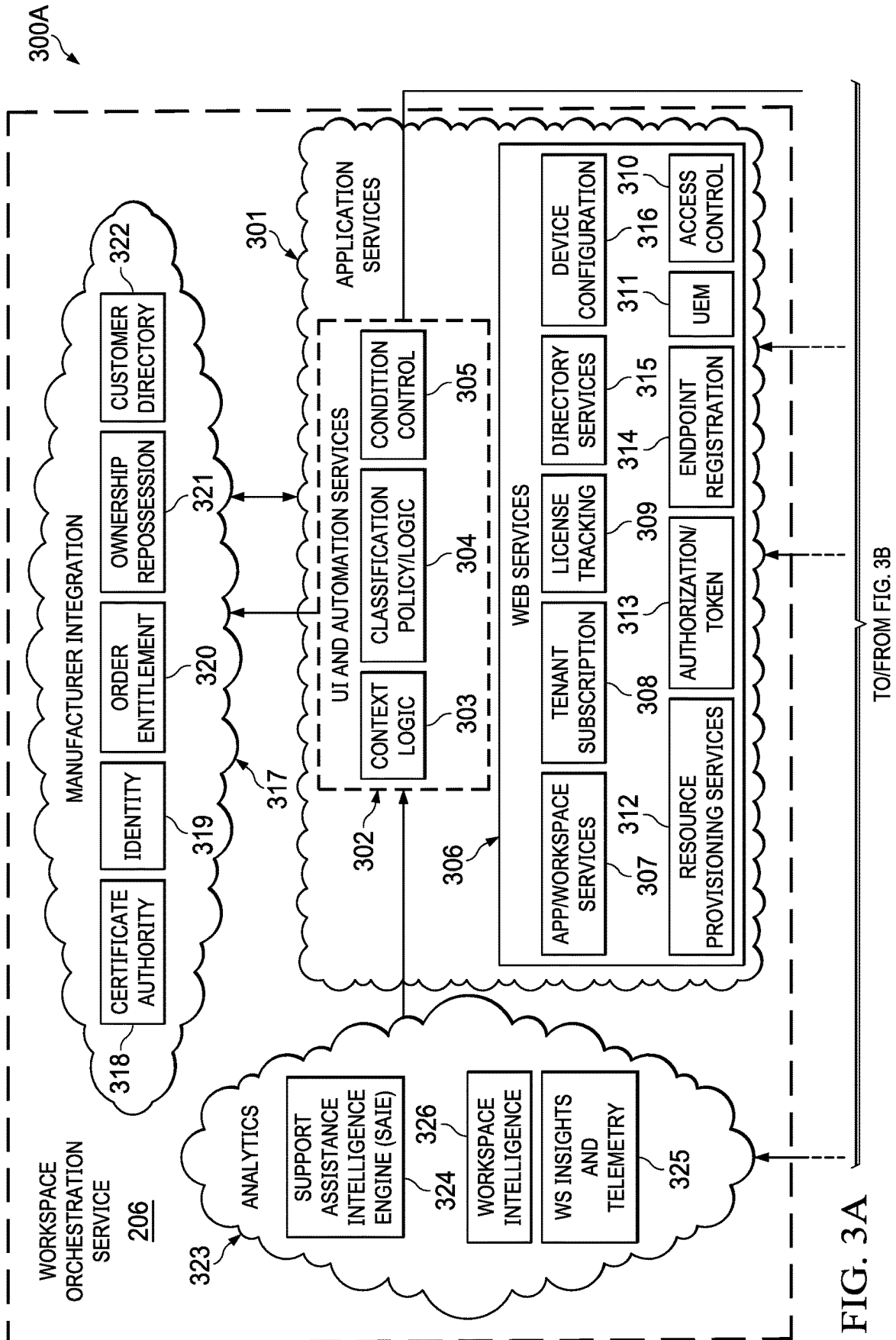


FIG. 1





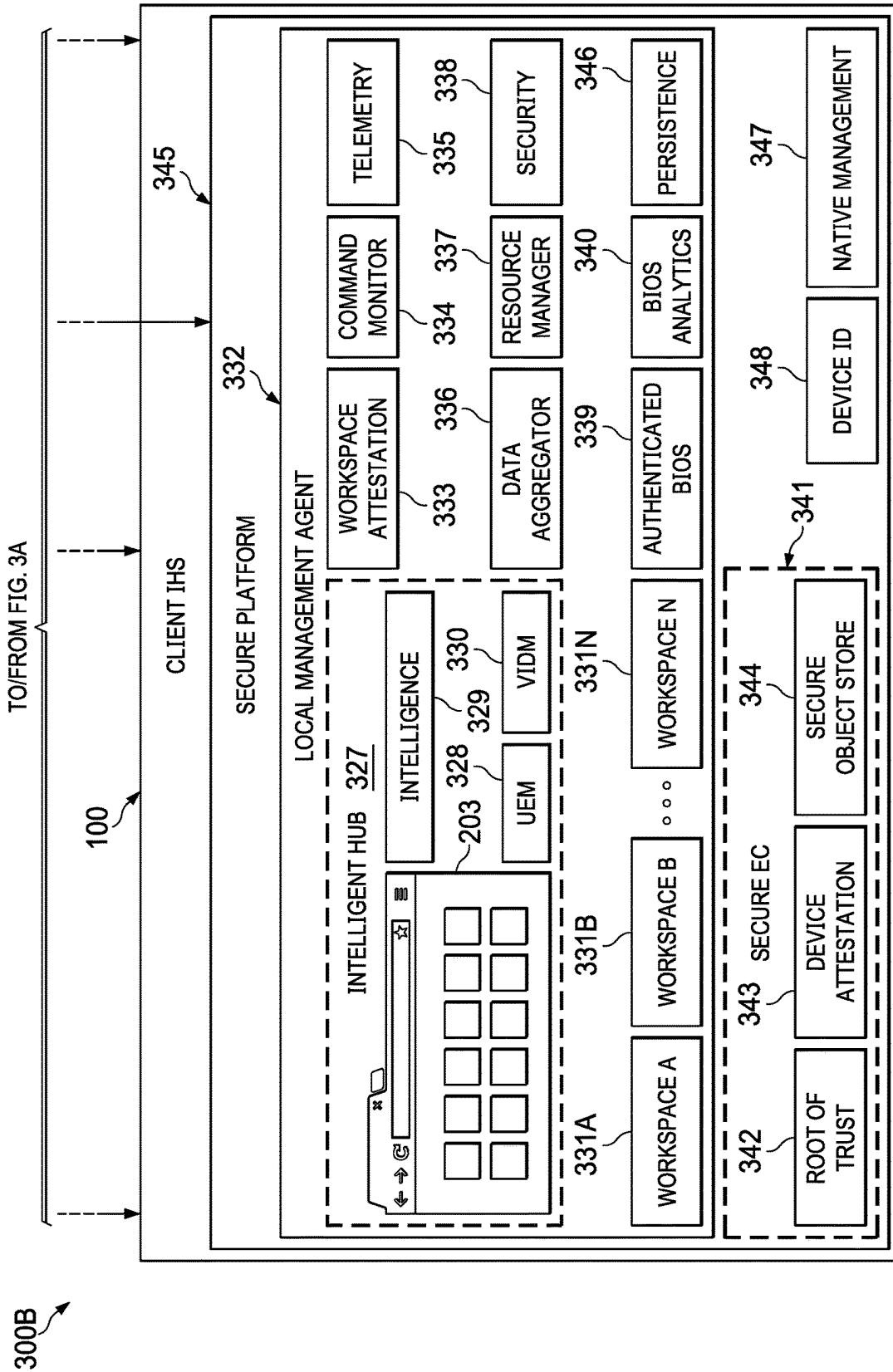


FIG. 3B

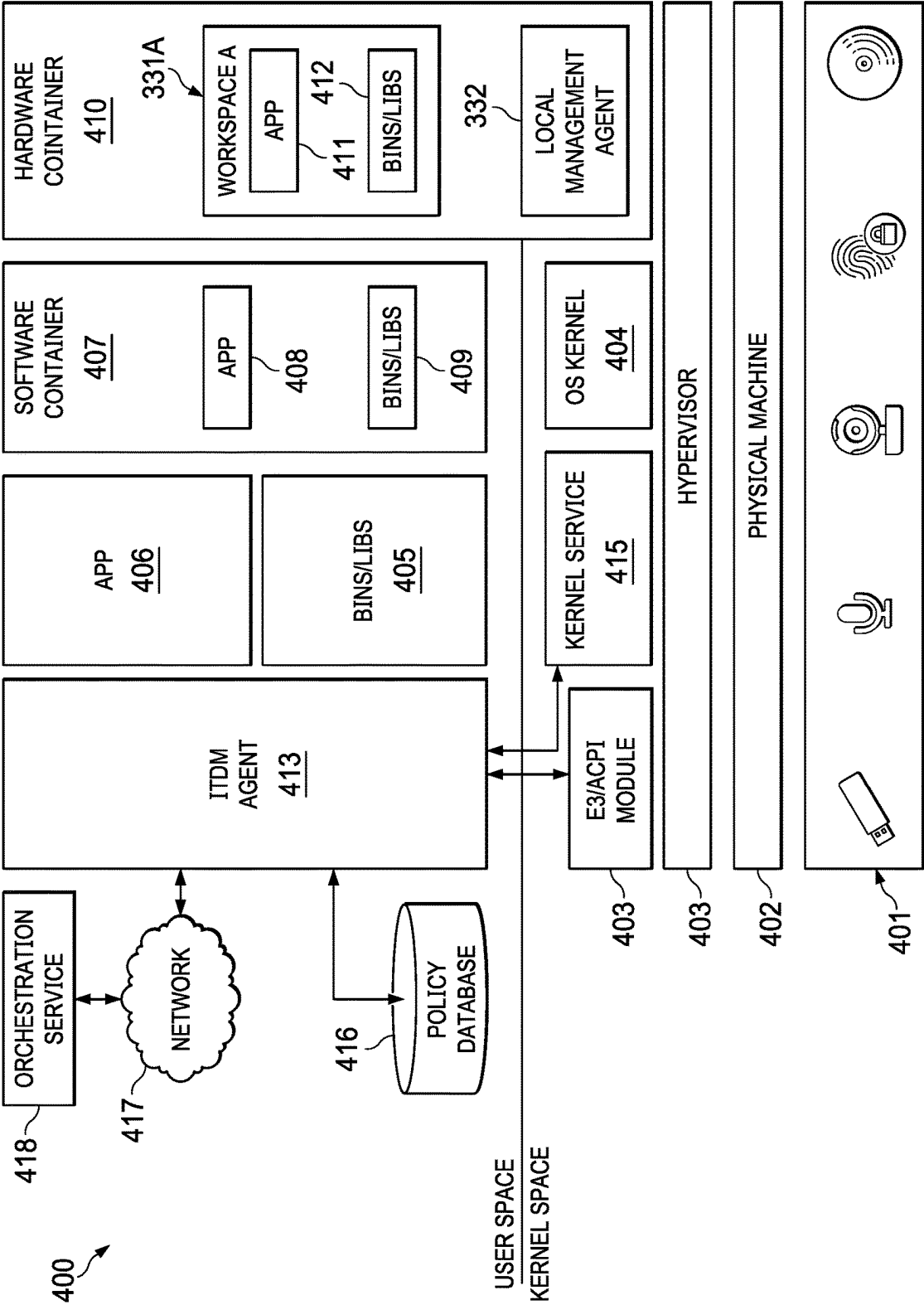
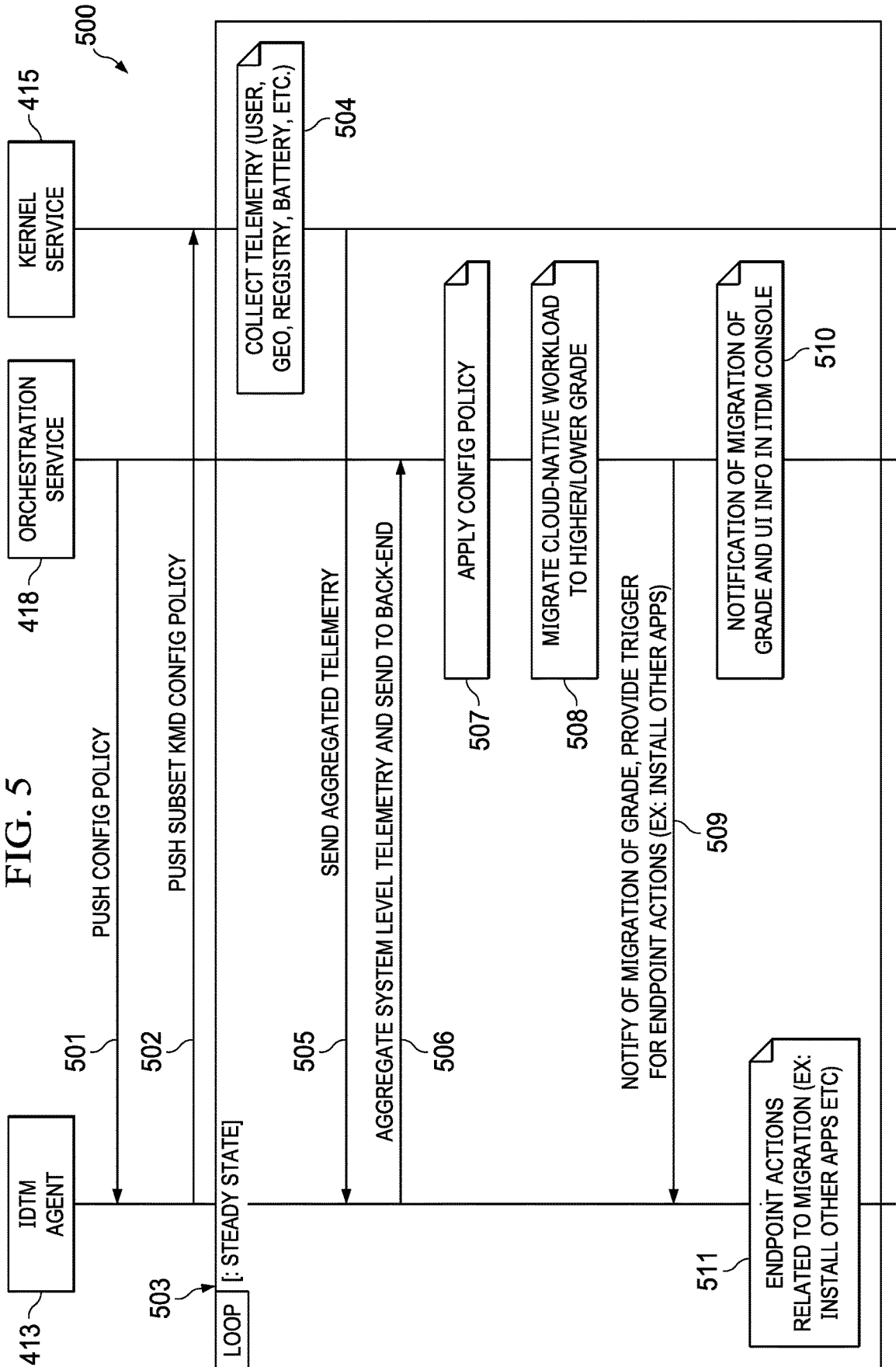


FIG. 4

FIG. 5



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MIGRATION OF WORKLOADS ACROSS CLOUD SERVICES BASED UPON ENDPOINT PERFORMANCE

FIELD

This disclosure relates generally to Information Handling Systems (IHSs), and, more specifically, to systems and methods for migration of workloads across cloud services based upon endpoint performance.

BACKGROUND

As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store it. One option available to users is an Information Handling System (IHS). An IHS generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information. Because technology and information handling needs and requirements vary between different users or applications, IHSs may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated.

Variations in IHSs allow for IHSs to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, IHSs may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

SUMMARY

Systems and methods for migration of workloads across cloud services based upon endpoint performance are described. In an illustrative, non-limiting embodiment, an Information Handling System (IHS) may include a processor and a memory coupled to the processor, the memory having program instructions stored thereon that, upon execution, cause the IHS to receive telemetry data indicative of a utilization of the IHS during execution of a workload by a first cloud service; determine a difference between a capability of the IHS and the utilization; and at least one of: (a) in response to the difference being greater than a first threshold, migrate the workload to a second cloud-based service with higher performance than the first cloud service; or (b) in response to the difference being smaller than a second threshold, migrate the workload to a third cloud-based service with lower performance than the first cloud service.

The telemetry data may include information indicative of at least one of: a software version, a driver installation, a registry entry, or a user setting. Additionally, or alternatively, the telemetry data may include information indicative of usage of a hardware resource accessed by the workload. For example, the hardware resource may include a battery. Additionally, or alternatively, the hardware resource may include at least one of: a storage, a memory, a keyboard, a mouse, a camera, or a display.

Additionally, or alternatively, the telemetry data may include information indicative of a display setting. Additionally, or alternatively, the telemetry data may include

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information indicative of a temporary storage location where a user stores a file used by the workload. Additionally, or alternatively, the telemetry data may include information indicative of at least one of: user proximity, IHS location, or IHS posture. Additionally, or alternatively, the telemetry data may include information indicative of at least one of: a network connection or a network activity.

Additionally, or alternatively, the telemetry data may include information indicative of one or more workspace settings applied upon instantiation of a workspace configured to execute the workload.

The workspace settings may include an indication of whether the workspace is a software-based or hardware-based. Additionally, or alternatively, the telemetry data may include information indicative of at least one of: a processor utilization or a memory utilization.

The program instructions, upon execution, may cause the IHS to receive the telemetry data from at least one of: an Energy Estimation Engine (E3) or an Advanced Configuration and Power Interface (ACPI) module. To determine the difference, the program instructions, upon execution, may cause the IHS to execute a Machine Learning (ML) model.

At least one of the first or second thresholds may be selected by a migration policy. The program instructions, upon execution, may cause the IHS to identify the migration policy based, at least, in part, upon a workspace definition associated with a workspace configured to execute the workload. The first, second, and third cloud services may be provided by a single cloud service provider. Alternatively, at least one of the first, second, and third cloud services may be provided by a different cloud service provider.

In another illustrative, non-limiting embodiment, a memory storage device may have program instructions stored thereon that, upon execution by an IHS, cause the IHS to: identify a difference between a capability of the IHS and a utilization of the IHS during execution of a workload using a cloud service tier and, in response to a change in the difference, migrate the workload to another cloud service tier with a different level of performance than the cloud service tier. In yet another illustrative, non-limiting embodiment, a method may include identifying a difference between a capability of an IHS and a utilization of the IHS during execution of a workload using a cloud service tier and, in response to a change in the difference, migrating the workload to another cloud service tier with a different level of performance than the cloud service tier.

BRIEF DESCRIPTION OF THE DRAWINGS

The present invention(s) is/are illustrated by way of example and is/are not limited by the accompanying figures, in which like references indicate similar elements. Elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale.

FIG. 1 is a diagram depicting examples of components of an Information Handling System (IHS) configured to modernize workspace and hardware lifecycle management in an enterprise productivity ecosystem, according to various embodiments.

FIG. 2 is a diagram depicting an example of a method for modernizing workspace and hardware lifecycle management in an enterprise productivity ecosystem, according to various embodiments.

FIGS. 3A and 3B are a diagram depicting an example of a system configured to modernize workspace and hardware lifecycle management in an enterprise productivity ecosystem, according to various embodiments.

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FIG. 4 is a diagram of an example of a heterogeneous workload environment configured to perform the migration of workloads across cloud services based upon IHS performance, according to various embodiments.

FIG. 5 is a flowchart of an example of a method for migration of workloads across cloud services based upon IHS performance, according to various embodiments.

DETAILED DESCRIPTION

For purposes of this disclosure, an IHS may include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an IHS may be a personal computer (e.g., desktop or laptop), tablet computer, mobile device (e.g., Personal Digital Assistant (PDA) or smart phone), server (e.g., blade server or rack server), a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. An example of an IHS is described in more detail below. FIG. 1 shows various internal components of an IHS configured to implement certain of the described embodiments. It should be appreciated that although certain embodiments described herein may be discussed in the context of a personal computing device, other embodiments may utilize various other types of IHSs.

FIG. 1 is a diagram depicting components of an example IHS 100 configured to modernize workspace and hardware lifecycle management in an enterprise productivity ecosystem, as well as to enable systems and methods for performance-based migration of workloads across cloud services. In some embodiments, IHS 100 may be employed to instantiate, manage, and/or terminate a workspace, such as a secure environment that may provide the user of IHS 100 with access to enterprise data while isolating the enterprise data from an Operating System (OS) and/or other applications executed by IHS 100.

As shown in FIG. 1, IHS 100 includes one or more processor(s) 101, such as a Central Processing Unit (CPU), operable to execute code retrieved from system memory 105. Although IHS 100 is illustrated with a single processor, other embodiments may include two or more processors, that may each be configured identically, or to provide specialized processing functions. Processor(s) 101 may include any processor capable of executing program instructions, such as an INTEL PENTIUM series processor or any general-purpose or embedded processors implementing any of a variety of Instruction Set Architectures (ISAs), such as the x86, POWERPC®, ARM®, SPARC®, or MIPS® ISAs, or any other suitable ISA. In the embodiment of FIG. 1, processor(s) 101 includes an integrated memory controller 118 that may be implemented directly within the circuitry of processor(s) 101, or memory controller 118 may be a separate integrated circuit that is located on the same die as processor(s) 101. Memory controller 118 may be configured to manage the transfer of data to and from system memory 105 of IHS 100 via high-speed memory interface 104.

System memory 105 that is coupled to processor(s) 101 via memory bus 104 provides processor(s) 101 with a high-speed memory that may be used in the execution of computer program instructions by processor(s) 101. Accordingly, system memory 105 may include memory components, such as static RAM (SRAM), dynamic RAM (DRAM), NAND Flash memory, suitable for supporting

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high-speed memory operations by processor(s) 101. In some embodiments, system memory 105 may combine both persistent, non-volatile memory and volatile memory.

In certain embodiments, system memory 105 includes secure storage 120 that may be a portion of the system memory designated for storage of information, such as access policies, component signatures, encryption keys, and other cryptographic information, for use in hosting a secure workspace on IHS 100. In such embodiments, a signature may be calculated based on the contents of secure storage 120 and stored as a reference signature. The integrity of the data stored in secure storage 120 may then be validated at a later time by recalculating this signature of the contents of the secure storage and comparing the recalculated signature against the reference signature.

IHS 100 utilizes chipset 103 that may include one or more integrated circuits that are coupled to processor(s) 101. In the embodiment of FIG. 1, processor(s) 101 is depicted as a component of chipset 103. In other embodiments, all of chipset 103, or portions of chipset 103 may be implemented directly within the integrated circuitry of processor(s) 101. Chipset 103 provides processor(s) 101 with access to a variety of resources accessible via bus 102. In IHS 100, bus 102 is illustrated as a single element. However, other implementations may utilize any number of buses to provide the illustrated pathways served by bus 102.

As illustrated, a variety of resources may be coupled to processor(s) 101 of IHS 100 through chipset 103. For instance, chipset 103 may be coupled to network interface 109, such as provided by a Network Interface Controller (NIC) that is coupled to IHS 100 and allows IHS 100 to communicate via a network, such as the Internet or a LAN. Network interface device 109 may provide IHS 100 with wired and/or wireless network connections via a variety of network technologies, such as wireless cellular or mobile networks (CDMA, TDMA, LTE etc.), WIFI and BLUETOOTH. In certain embodiments, network interface 109 may support connections between a trusted IHS component, such as trusted controller 115, and a remote orchestration service. In such embodiments, a connection supported by network interface 109 between the remote orchestration service and the trusted component may be considered an out-of-band (OOB) connection that is isolated from the OS of the IHS.

Chipset 102 may also provide access to one or more display device(s) 108 via graphics processor 107. In certain embodiments, graphics processor 107 may be comprised within one or more video or graphics cards or an embedded controller installed as components of IHS 100. Graphics processor 107 may generate display information and provide the generated information to one or more display device(s) 108 coupled to IHS 100, where display device(s) 108 may include integrated display devices and/or external display devices coupled to IHS, such as via an I/O port 116, where display device(s) 108 may include integrated display devices and/or external display devices coupled to IHS. In certain embodiments, graphics processor 107 may be integrated within processor 101. The one or more display devices 108 coupled to IHS 100 may utilize LCD, LED, OLED, or other thin film display technologies. Each display device 108 may be capable of touch input such as via a touch controller that may be an embedded component of display device 108, graphics processor 107, or a separate component of IHS 100 accessed via bus 102.

In certain embodiments, chipset 103 may utilize one or more I/O controllers to access hardware components such as user input devices 111 and sensors 112. For instance, I/O

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controller 110 may provide access to user-input devices 110 such as a keyboard, mouse, touchpad, touchscreen and/or other peripheral input devices. User input devices 111 may interface with I/O controller 110 through wired or wireless connections. Sensors 112 accessed via I/O controllers 110 may provide access to data describing environmental and operating conditions of IHS 100 (e.g., accelerometers, gyroscopes, hinge sensors, rotation sensors, hall effect sensors, temperature sensors, voltage sensors, sensors, IR sensors, photosensors, proximity sensors, distance sensors, magnetic sensors, microphones, ultrasonic sensors, etc.).

In some cases, chipset 103 may include a sensor hub capable of utilizing information collected by sensors 112 in determining the relative orientation and movement of IHS 100. For instance, the sensor hub may utilize inertial movement sensors, that may include accelerometer, gyroscope, and magnetometer sensors, and are capable of determining the orientation and movement of IHS 100 (e.g., IHS 100 is motionless on a relatively flat surface, IHS 100 is being moved irregularly and is likely in transport, the hinge of IHS 100 is oriented in a vertical direction).

In certain embodiments, the sensor hub may also include capabilities for determining a location and movement of IHS 100 based on triangulation of network signal and based on network information provided by the OS or network interface 109. In some embodiments, the sensor hub may support additional sensors, such as optical, infrared and sonar sensors, that may provide support for xR (virtual, augmented, and/or mixed reality) sessions hosted by the IHS 100 and may be used by the sensor hub provide an indication of a user's presence near IHS 100, such as whether a user is present, absent, and/or facing integrated display 108.

In cases where the end-user is present before IHS 100, the sensor hub may further determine a distance of the end-user from the IHS, where this determination may be made continuously, at periodic intervals, or upon request. The detected or calculated distances may be used by processor 101 to classify the user as being in the IHS's near-field (user's position < threshold distance A), mid-field (threshold distance A < user's position < threshold distance B, where B > A), or far-field (user's position > threshold distance C, where C > B). As described in additional detail below, the failure to detect an authenticated user of IHS 100 within a proximity of IHS 100 may result in a change in the security profile of IHS 100, thus triggering a re-evaluation of the security risk of workspaces operating on IHS 100. Similar re-evaluation may be triggered based on the detection of additional individuals in proximity to IHS 100.

In embodiments where IHS 100 may support multiple physical configurations, such as a convertible laptop, N-in-1 device, or the like, the sensor hub may utilize one or more mode sensors 112 that collect readings that may be used in determining the posture in which IHS 100 is physically configured. In certain embodiments, such posture determinations may be additionally made using the movement and orientation information provided by sensors 112. In laptop and convertible laptop embodiments, for example, processor 101 or trusted controller 115 may utilize a lid position sensor 112 to determine the relative angle between the two panels of the laptop in order to determine the mode in which IHS 100 is physically configured. In such embodiments, the lid position sensor may measure the angle of rotation of the hinge that connects the base panel and lid panel of IHS 100. In some embodiments, processor 101 or trusted controller 115 may provide collected lid position information, such as the hinge angle, to the sensor hub for use in determining the posture in which IHS 100 is configured. In some embodi-

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ments, the sensor hub may interface directly with the lid position sensor in determining hinge angle information.

The sensor hub may determine the posture of IHS 100 based, at least in part, on the angle of rotation of the hinge of IHS 100 from a closed position. A first range of hinge angles from a closed position may indicate a laptop posture, a second range of hinge angles may indicate a landscape posture and a third range of angles may indicate a tablet posture. The sensor hub may additionally utilize orientation and movement information collected from inertial movement sensors 112 to further determine the posture in which IHS 100 is physically configured. For instance, if the sensor hub determines that IHS 100 is configured with a hinge angle of a laptop configuration, but IHS 100 is oriented on its side, the IHS may be determined to be in a book mode. If IHS 100 is determined to be tilted such that the hinge is oriented between horizontal and vertical, the user's face is detected to be facing the integrated display, and IHS 100 is experiencing slight movement, the sensor hub may determine that IHS 100 is being used in a book posture. The sensor hub may determine that IHS 100 is opened to a 180-degree hinge angle and lies on a flat surface, thus indicating that IHS 100 it is being used in a landscape posture. The sensor hub may similarly determine that IHS 100 is in a tent configuration, in response to detecting a hinge angle within a defined range, such as between 300 and 345 degrees, and also detecting an orientation of IHS 100 where the hinge is aligned horizontally and is higher than both of the display panels of IHS 100.

Other components of IHS 100 may include one or more I/O ports 116 for communicating with peripheral external devices as well as various input and output devices. For instance, I/O 116 ports may include HDMI (High-Definition Multimedia Interface) ports for use in connecting external display devices to IHS 100 and USB (Universal Serial Bus) ports, by which a variety of external devices may be coupled to IHS 100. In some embodiments, external devices coupled to IHS 100 via an I/O port 116 may include storage devices that support transfer of data to and from system memory 105 and/or storage devices 119 of IHS 100. As described in additional detail below, the coupling of storage devices via an I/O port 116 may result in a change in the security profile of IHS 100, thus triggering a re-evaluation of the security risk of workspaces operating on IHS 100.

Chipset 103 also provides processor(s) 101 with access to one or more storage devices 119. In various embodiments, storage device 119 may be integral to IHS 100, or may be external to IHS 100. In certain embodiments, storage device 119 may be accessed via a storage controller that may be an integrated component of the storage device. Storage device 119 may be implemented using any memory technology allowing IHS 100 to store and retrieve data. For instance, storage device 119 may be a magnetic hard disk storage drive or a solid-state storage drive. In some embodiments, storage device 119 may be a system of storage devices, such as a cloud drive accessible via network interface 109.

As illustrated, IHS 100 also includes BIOS (Basic Input/Output System) 117 that may be stored in a non-volatile memory accessible by chipset 103 via bus 102. Upon powering or restarting IHS 100, processor(s) 101 may utilize BIOS 117 instructions to initialize and test hardware components coupled to IHS 100. BIOS 117 instructions may also load an OS for use by IHS 100. BIOS 117 provides an abstraction layer that allows the OS to interface with the hardware components of IHS 100. The Unified Extensible Firmware Interface (UEFI) was designed as a successor to BIOS. As a result, many modern IHSs utilize UEFI in

addition to or instead of a BIOS. As used herein, BIOS is intended to also encompass UEFI.

In the illustrated embodiment, BIOS 117 includes a pre-defined memory or memory region that may be referred to as NVM (Non-Volatile Memory) mailbox 106. In such an implementation, mailbox 106 may provide a secured storage location for use in storing workspace access policies, signatures, cryptographic keys or other data utilized to host and validate a workspace on IHS 100. In certain embodiments, BIOS mailbox 106 may be utilized as a secure storage utilized by a remote orchestration service in order to store access policies and cryptographic keys for use in delivering and deploying a secured container on IHS 100. BIOS mailbox 106 and secured storage 120 in system memory 105 may be utilized in this manner instead of, or in conjunction with, out-of-band functions implemented by trusted controller 115.

In certain embodiments, trusted controller 115 is coupled to IHS 100. For example, trusted controller 115 may be an embedded controller (EC) that is installed as a component of the motherboard of IHS 100. In various embodiments, trusted controller 115 may perform various operations in support of the delivery and deployment of a workspace to IHS 100. In certain embodiments, trusted controller 115 may interoperate with a remote orchestration service via an out-of-band communications pathway that is isolated from the OS that runs on IHS 100. Network interface 109 may support such out-of-band communications between trusted controller 115 and a remote orchestration service.

Trusted controller 115 may receive cryptographic information required for secure delivery and deployment of a workspace to IHS 100. In such embodiments, the cryptographic information may be stored to secured storage 121 maintained by trusted controller 115. Additionally, or alternatively, trusted controller 115 may support execution of a trusted operating environment that may support cryptographic operations used to deploy a workspace on IHS 100. Additionally, or alternatively, trusted controller 115 may support deployment of a workspace within the OS of IHS 100 via an out-of-band communications channel that is isolated from the OS and allows the workspace to communicate with a trusted agent process of the OS.

Trusted controller 115 may also provide support for certain cryptographic processing used to support secure deployment and operation of workspaces on IHS 100. In some embodiments, such cryptographic processing may be provided via operations of a secure operating environment hosted by trusted controller 115 in isolation from the software and other hardware components of IHS 100. In some embodiments, trusted controller 115 may rely on cryptographic processing provided by dedicated cryptographic hardware supported by the IHS, such as a TPM (Trusted Platform Module) microcontroller. In some embodiments, the secured storage 121 of trusted controller 115 may be utilized to store cryptographic information for use in authorization of workspaces.

In certain embodiments, trusted controller 115 may be additionally configured to calculate signatures that uniquely identify individual components of IHS 100. In such scenarios, trusted controller 115 may calculate a hash value based on the configuration of a hardware and/or software component coupled to IHS 100. For instance, trusted controller 115 may calculate a hash value based on all firmware and other code or settings stored in an onboard memory of a hardware component, such as a network interface 109. Such hash values may be calculated as part of a trusted

process of manufacturing IHS 100 and may be maintained in the secure storage 121 as a reference signature.

Trusted controller 115 may be further configured to recalculate a hash value at a later time for such a component. The hash value recalculated for the component may then be compared against the reference hash value signature in order to determine if any modifications have been made to a component, thus indicating the component has been compromised. In this manner, trusted controller 115 may be used to validate the integrity of hardware and software components installed on IHS 100. In certain embodiments, remote orchestration service 206 may verify the integrity of trusted controller 115 in the same manner, by calculating a signature of trusted controller 115 and comparing it to a reference signature calculated during a trusted process for manufacture of IHS 100. In various embodiments, one or more of these operations supported by trusted controller 115 may be implemented using BIOS 117.

Trusted controller 115 may also implement operations for interfacing with a power adapter in managing power for IHS 100. Such operations may be utilized to determine the power status of IHS 100, such as whether IHS 100 is operating from battery power or is plugged into an AC power source. Firmware instructions utilized by trusted controller 115 may be used to operate a secure execution environment that may include operations for providing various core functions of IHS 100, such as power management and management of certain operating modes of IHS 100 (e.g., turbo modes, maximum operating clock frequencies of certain components, etc.).

In managing operating modes of IHS 100, trusted controller 115 may implement operations for detecting certain changes to the physical configuration of IHS 100 and managing the modes corresponding to different physical configurations of IHS 100. For instance, where IHS 100 is a laptop computer or a convertible laptop computer, trusted controller 115 may receive inputs from a lid position sensor 112 that may detect whether the two sides of the laptop have been latched together to a closed position. In response to lid position sensor 112 detecting latching of the lid of IHS 100, trusted controller 115 may initiate operations for shutting down IHS 100 or placing IHS 100 in a low-power mode.

IHS 100 may support the use of various power modes. In some embodiments, the power modes of IHS 100 may be implemented through operations of trusted controller 115 and/or the OS of IHS 100. In various embodiments, IHS 100 may support various reduced power modes in order to reduce power consumption and/or conserve battery power when IHS 100 is not actively in use, and/or to control a level of performance available to the user by increasing or decreasing a maximum operating clock frequency of a component of IHS 100 (e.g., processor(s) 101).

In some embodiments, an IHS 100 may not include all of the components shown in FIGURE 1. In other embodiments, an IHS 100 may include other components in addition to those that are shown in FIG. 1. Furthermore, some components that are represented as separate components in FIG. 1 may instead be integrated with other components. For example, in certain embodiments, all or a portion of the operations executed by the illustrated components may instead be provided by components integrated into processor (s) 101 as a System-on-Chip.

In some embodiments, the construction of a workspace for a particular purpose and for use in a particular context may be orchestrated remotely from IHS 100 by workspace orchestration services 206, such as described with regard to FIG. 2. In some embodiments, portions of the workspace

orchestration may be performed locally on IHS 100. IHS 100 may be configured with program instructions that, upon execution, cause IHS 100 to perform one or more of the various operations disclosed herein. In some embodiments, IHS 100 may be an element of a larger enterprise system that may include any number of similarly configured IHSs in network communications with each other.

FIG. 2 is a diagram depicting an example of method 200 for securing a dynamic workspace in an enterprise productivity ecosystem. For sake of illustration, method 200 has been split into three phases: workspace initialization phase 200A, workspace orchestration phase 200B, and workspace termination phase 200C. During initialization 200A, user 201 (e.g., an enterprise user) operates an IHS 100 (e.g., a desktop, a laptop, a tablet, a smart phone, etc.) such as described with regard to FIG. 1 within physical environment 202 (e.g., any type of environment and its associated context, including physical location, geographic location, location within a particular facility or building, detected networks, time of day, proximity of the user, individuals in the vicinity of IHS 100, etc.).

Method 200 starts with an action by user 201 at a launch point 203 that may be, for example, a corporate launch point provided by an employer of user 201, a launch point 203 provided by the manufacturer of IHS 100, or a launch point provided as a service to user 201 by a third-party. Particularly, user 201 operates IHS 100 to access launch point 203 that is provided, for example, in the form of a web portal, a portal application running in the OS of IHS 100, a special-purpose portal workspace operating on IHS 100, or the like. In various implementations, launch point 203 may include Graphical User Interface (GUI) elements representing different software applications, data sources and/or other resources that the user may desire to execute and/or manipulate. In various embodiments, launch point may provide a graphical, textual and/or audio interface by which data or other resource may be requested by a user 201. As such, authenticated user 201 may be provided a launch point that provides visibility as to one or more software applications and an aggregation of user's data sources available across all of their datastores (e.g., local storage, cloud storage, etc.).

As described in additional detail below, workspaces for providing user 201 with access to requested data or other resources may operate using a local management agent 332 that operates on IHS 100 and is configured to interoperate with workspace orchestration service 206. In various embodiments, launch point 203 may be provided in the form of a portal (e.g., a webpage, OS application or special purpose workspace) that allows user 201 to request access to managed resources. In various embodiments, launch point 203 may be hosted by remote workspace orchestration service 206, local management agent 332 on IHS 100, or any suitable combination thereof. Examples of launch point 203 technologies may include WORKSPACE ONE INTELLIGENT HUB from WMWARE, INC., and DELL HYBRID CLIENT from DELL TECHNOLOGIES INC., among others.

Initialization phase 200A begins when user 201 chooses to launch an application or access a data source managed by workspace orchestration service 206. In response to an access request issued by user 201 (e.g., the user "clicks" on an icon of launch point 203), local management agent 332 of IHS 100 collects initial security and productivity context information at 204. For example, security context information may include attributes indicating a security risk associated with: the data and/or application being requested, a level of risk presented by the user 201, the hardware utilized

by IHS 100, the logical environment of IHS 100 in which a workspace will be deployed to provide access to the requested data and/or application, and the physical environment 202 in which IHS 100 is currently located.

Accordingly, in this disclosure, the term "security context" generally refers to data or other information related to a security posture in which a workspace will be deployed and utilized, where the security posture may be based on the user, IHS 100, data to be accessed via the workspace, and/or environment 202. A security context may be quantified as a security risk score in support of evaluations of the level or risk associated with providing user 201 access to requested data and/or application while using IHS 100 in the particular context. A "security risk score" generally refers to a numerical value usable to score, quantify, or measure various security characteristics of the security context associated with a request. A risk score may be an aggregate score associated with the overall security risk context, whereas a "risk metric" may be a measurement of risk for a sub-category of some part of the security context.

For example, security metrics that may be used in the calculation of a security risk score for a particular security context may include, but are not limited to: a classification of the requested data source and/or application, authentication factors used to identify user 201, the location of IHS 100, a role or other group classifications associated with user 201, validation of networks in use by IHS 100, type of network in use by IHS 100, network firewall configurations in use by IHS 100, indicators of attack (IoA), indicators of compromise (IoC) regarding IHS 100 or a resource being requested by user 201, patch levels associated with the OS and other applications in use on IHS 100, availability of encryption, type of available encryption, access to secured storage, use of attestable hardware by IHS 100, supported degree of workspace isolation by IHS 100, etc.

The term "productivity context" generally refers to user productivity associated with a workspace, user, IHS, or environment. A "productivity score" generally refers to an index usable to score, quantify, or measure various productivity characteristics of a productivity context.

Examples of productivity context information include, but are not limited to: the hardware of the IHS, the software of the IHS, including the OS, power states and maximum clock frequencies of selected components of the IHS, peripheral devices coupled to the IHS, either permanently or temporarily, networks available to the IHS and the performance characteristics of those networks, software installers available on the IHS, etc.

Initial productivity and security targets for a workspace may be calculated based on the context of user's 201 actions combined with the productivity and security context in which the workspace will operate. The productivity and security targets may also be based on user's 201 behavioral analytics, IHS 100 telemetry and/or environmental information (e.g., collected via sensors 112). In some cases, at 205, a local management agent operating on IHS 100 may calculate initial security and productivity targets based upon the collected security and productivity context. In other cases, remote workspace orchestration service 206 may calculate security and productivity targets.

As used herein, the term "security target" generally refers to the attack surface presented by a workspace that is created and operated based on a workspace definition, while the term "productivity target" generally refers to the productivity characteristics of a particular workspace definition. Examples of a productivity target include, but are not limited to: type of data or data source available to user 201,

minimum latency of a workspace, etc. Conversely, attributes that may be used to characterize a security target may include, but are not limited to: a minimum security score for a workspace, a minimum trust score of IHS 100, authentication requirements for user 201 (e.g., how many authentication factors are required, frequency of re-authentication), minimum level of trust in the network utilized by a workspace, required isolation of a workspace from IHS 100, the ability to access browser within a workspace, the ability to transfer data between workspaces, the ability to extend a workspace, etc.

Moreover, the term “workspace definition” generally refers to a collection of attributes that describe aspects of a workspace that may be assembled, created, and deployed in a manner that satisfies a security target (i.e., the definition presents an attack surface that presents an acceptable level of risk) and a productivity target (e.g., data access, access requirements, upper limits on latency, etc.) in light of the security context (e.g., location, patch level, threat information, network connectivity, etc.) and the productivity context (e.g., available device type and performance, network speed, etc.) in which the workspace is to be deployed. A workspace definition may enable fluidity of migration of an instantiated workspace, since the definition supports the ability for a workspace to be assembled on any target OS or IHS that is configured for operation with the workspace orchestration service 206.

In describing capabilities and constraints of a workspace, a workspace definition 208 may prescribe one or more of: authentication requirements for user 201, containment and/or isolation of the workspace (e.g., local application, sandbox, docker container, progressive web application or “PWA,” Virtual Desktop Infrastructure “VDI,” etc.), primary applications that can be executed in the defined containment of the workspace to enable user 201 to be productive with one or more data sources, additional applications that enhance productivity, security components that reduce the scope of the security target presented by the productivity environment (DELL DATA GUARDIAN from DELL TECHNOLOGIES INC., an anti-virus, etc.), the data sources to be accessed and requirements for routing that data to and from the workspace containment (e.g., use of VPN, minimum encryption strength), workspace capabilities to independently attach other resources; etc.

In some implementations, workspace definitions may be based at least in part on static policies or rules defined, for example, by an enterprise’s Information Technology (IT) Decision Maker (ITDM). In some implementations, static rules may be combined and improved upon by machine learning (ML) and/or artificial intelligence (AI) algorithms that evaluate historical productivity and security data collected as workspaces are life cycled. In this manner, rules may be dynamically modified over time to generate improved workspace definitions. If it is determined, for instance, that a user dynamically adds a text editor every time he uses MICROSOFT VISUAL STUDIO from MICROSOFT CORPORATION, then workspace orchestration service 206 may autonomously add that application to the default workspace definition for that user.

Still with respect to FIG. 2, during orchestration 200B, the initial security and productivity targets are processed and/or reconciled against resources, device capabilities, and cloud services available, etc., to produce a workspace definition at 208. As described, a workspace definition may specify capabilities and constraints of a workspace, such as: runtime security requirements of the workspace containment (e.g., such as isolation from the OS of IHS 100 or from certain

hardware of IHS 100), the use of reference measurements to attest to the integrity of the workspace once running, applications to be provided for operation within the workspace, aggregation of resources available via the workspace, access configurations (e.g., virtual private network or “VPN”), etc.

The initial workspace definition may then be utilized by automation engine 302 of workspace orchestration service 206 to coordinate assembly 209 and instantiation 210 of a workspace on an appropriate platform—e.g., on the cloud or on IHS 201—based on the security and productivity contexts in which the workspace will operate. In cases where a workspace is cloud-hosted, automation engine 302 may assemble and instantiate a remote workspace that may be accessed via a secure connection established via a web browser or other web-based component operating on IHS 100. In some embodiments, automation engine 302 may resolve configuration conflicts between a workspace definition and the user’s inputs in the operation of a workspace.

The instantiated workspace is operated by user 201 at 211, and new productivity and security context information related to the behavior or use of data is generated at 212. This operation of a workspace may result in a change or new classification of data based upon what user 201 has done, accessed, and/or created, thus resulting in a change to the security context of the workspace. To the extent the user’s behavioral analytics, device telemetry, and/or the environment has changed to a quantifiable degree, these changes in security context may serve as additional input for a reevaluation of the security and performance targets at 207 by automation engine 302. Additionally, or alternatively, new workspace context, security target, and/or productivity target may be now measured against the initial targets, and the result may cause automation engine 302 to produce a new workspace definition at 208, if appropriate.

Particularly, if an instantiated workspace has parameters that fall outside of the range of the target indexes such that a difference between additional or updated context information and the initial or previous context information is scored below a threshold value, automation engine 302 may process the assembly of modifications to an existing workspace and deploy such modifications at 210. Conversely, if the difference between the additional or updated context information and the initial or previous context information is scored above a threshold value, automation engine 302 may generate a new workspace at 210. Session data metadata and context may be preserved by data aggregation engine 336, and session data may be restored as applicable.

Additionally, or alternatively, method 200 may terminate or retire the initial or previous workspace at 213, as part of termination phase 200C. In some cases, user action may initiate the termination process (e.g., user 201 closes application or browser accessing data) and/or termination may take place automatically as part of an adjustment in workspace definition (e.g., the isolated environment is instructed to terminate by automation engine 302). Still as part of termination phase 200C, workspace resources of IHS 100 and/or at workspace orchestration service 206 may be released.

As such, in various embodiments, method 200 enables secure user productivity even when a workspace operates on an IHS or cloud platform that is not under direct management. Method 200 also provides for dynamic or adaptive configurations and policies allowing for the best possible user experience while maintaining appropriate level of security. In some cases, the definition of a productivity environment and access requirements may be selected based upon productivity and security dependencies and targets, and the

definition of capabilities related to the workspace may be adaptive in nature. Particularly, workspace definition attributes may be dynamically selected based upon historical productivity and security information, based upon each individual user or group's behavior.

FIGS. 3A and 3B show a diagram of an example of system components **300A** and **300B** (collectively referred to as "system **300**") configured to modernize workspace and hardware lifecycle management in an enterprise productivity ecosystem. Particularly, system **300** may include one or more IHSs remotely located and/or networked having program instructions stored thereon that, upon execution, cause the one or more IHSs to perform various workspace orchestration operations described herein, including, but not limited to: the dynamic evaluation of security and productivity targets based upon updated context information received from IHS **100**, the calculation of risk scores and other productivity and security metrics based on ongoing collection of context information, the generation of workspace definitions, and the assembly of one or more files or policies that enable the instantiation of a workspace in accordance with a workspace definition at a cloud service and/or IHS **100**.

System **300** may include program instructions that, upon execution, cause IHS **100** to perform various local management operations described herein, including, but not limited to, the collection of productivity and security context information, the calculation of productivity scores and/or risk scores, the instantiation, execution, and modification of a workspace based upon files, definitions, or policies, such as workspace definitions.

Components **300A** and **300B** of system **300** may be coupled to and/or in communication with each other via any suitable network technology and/or protocol, which allows workspace orchestration service **206** to be remotely provided with respect to local management agent **332**. As described with regard to FIG. 1, an IHS according to embodiments may include a component such as a trusted controller that may support certain secure out-of-band communications that are independent from the OS of IHS **100**. In some embodiments, such a trusted controller may be configured to support deployment and operation of local management agent **332** and/or to report changes in context to workspace orchestration service **206**.

As illustrated in system component **300A** of FIG. 3A, workspace orchestration service **206** may include a number of sub-components that support deployment and ongoing evaluation and adaptation of workspaces on IHS **100**. Embodiments of workspace orchestration service **206** may include systems that may support: web services **306**, manufacturer integration **317**, and analytics **323**. Moreover, web services **306** may comprise application services **301** and user interface (UI) and automation services **302**.

Analytics services **323** may be configured to receive and process context information from IHS **100**, both during initial configuration of a workspace and in ongoing support of workspaces, and to provide that information, along with any analytics generated, to context logic **303** of application services **301**. Based on information collected during the deployment and ongoing support of workspaces, support assistance intelligence engine (SAIE) **324** may be configured to generate and/or analyze technical support information (e.g., updates, errors, support logs, etc.) for use in diagnosing and repairing workspace issues. Workspace insights and telemetry engine **325** may be configured to analyze and/or produce device-centric, historical, and behavior-based data (e.g., hardware measurements, use of

features, settings, etc.) resulting from the operation of workspaces. Workspace intelligence module **326** may include any suitable intelligence engine for processing and evaluating context data in order to identify patterns and tendencies in the operation of workspaces and in the adaptation of workspaces based on context changes.

Application services **306** system of workspace orchestration service **206** includes UI and automation services **302** system that may include context logic or engine **303**, classification policy **304**, and condition control module or engine **305**. Context logic or engine **303** may support processing of context information in making risk assessments (e.g., evaluating the risk associated requests by the user against the context of the user's behavior, history of the user's IHS, capabilities of the user's IHS, and environmental conditions). For instance, security context information collected by IHS **100** may be provided to workspace orchestration service **206** where it may be used, such as by context logic **303**, to calculate a risk score associated with a request for use of a managed data source and/or application. Classification policy **304** may include administrator and machine-learning defined policies describing risk classifications associated with different security contexts, such as risk classifications for specific data, locations, environments, IHSs, logical environments, or user actions (e.g., use of high-risk data requires use of a workspace definition suitable for use with a risk score above a specific value). Condition control module or engine **305** may include intelligence providing automated decision making for appropriately aligning risk and context. In some cases, condition control module or engine **305** may dynamically deploy a solution to address any detected misalignment of risk and context. For instance, upon requesting access to a highly classified data source that results in a significant increase in risk score, the condition control engine may select workspace definition modifications that implement security procedures that are suitable for the higher risk score.

Application services **301** may include a group of web services **306** called on by UI and automation services **302** to support various aspects of the orchestration of workspaces. Particularly, web services **306** may include application and workspace services **307** that may assemble and package applications for deployment in a workspace (e.g., an ".msix" file packaged and deployed to a MICROSOFT HYPER-V container). In some embodiments, a workspace definition may be used to specify whether a user will be provided access to an application in this manner. Web services **306** may also include a tenant subscription module **308**, that performs dynamic configuration of an IHS and deployment of the described workspace orchestration services at the point-of-sale (POS) of an IHS. A license tracking module **309** may be used to maintain and track license information for software, services, and IHSs. An access control module **310** may provide top level access controls used in controlling access to data and applications by authorized users. A Unified Endpoint Management (UEM) module **311** may be configured to support the described orchestration of workspaces on various different IHSs that may be utilized by a particular user.

Web services **306** that may be used in support of workspaces may further include resource provisioning services **312** for configuring an IHS or workspace with secrets/credentials necessary to access specific resources (e.g., credentials for use of VPNs, networks, data storage repositories, workspace encryption, workspace attestation, and workspace-to-device anchoring). In some cases, resource provisioning services **312** may include secrets provisioned

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as part of a trusted assembly process of IHS 100 and, in some instances, associated with a unique identifier 348 of the IHS 100. Web services 306 may also include an authorization/token module that provides identity functions and may connect to various authentication sources, such as, for example, Active Directory. Endpoint registration module 314 may be configured to register IHSs and/or workspaces with management service that tracks the use of the described workspace orchestration. In some scenarios, a directory services 315 module may be configured to provide active directory services (e.g., AZURE ACTIVE DIRECTORY from MICROSOFT). Device configuration services 316 enable central configuration, monitoring, managing, and optimization of workspaces that in certain contexts may operate remotely from an IHS and may only present the user of the IHS with an image of the workspace output. In cooperation with resource provisioning services 312, device configuration services 316 may also handle secret creation and IHS configuration, and in some cases, may be out-of-band capable and handle selected operations to the endpoint.

Still referring to FIG. 3A, manufacturer integration components 317 communicate with application services 301 and client IHS 100 to provide features that are usable during workspace evaluation and instantiation, where these features are based upon information available to the manufacturer of client IHS 100. For instance, certificate authority 318 may include an entity that issues digital certificates that may be used in validating the authenticity and integrity of the hardware of IHS 100. Identity service module or engine 319 may be configured to manage the user's or owner's identity as well as brokering identification for use of customer directory 322.

Order entitlement module or engine 320 may be responsible for managing the entitlements purchased as well as the associated issued certificates signed by 318. Ownership repository 321 may manage user entitlements associated with IHSs and their ownership and may provide support for users transferring ownership of an IHS and conveying the entitlements associated with that IHS. In certain scenarios, ownership repository 321 may use this transfer of ownership to decommission the secrets associated with the entitlements embedded in the IHS. Customer directory 322 may be configured to authenticate and authorize all users and IHSs in a network, such as assigning and enforcing security policies for all IHSs and installing or updating software (in some cases, customer directory 322 may work in cooperation and/or may be the same as directory services 315).

Referring now to IHS 100 of FIG. 3B, in some embodiments, IHS 100 may be configured to operate local management agent 332 that may run within a secure execution environment 345 hosted by trusted controller 341, such as trusted controller 115 of FIG. 1. In other embodiments, local management agent 332 may operate as a trusted and attestable process of the OS of IHS 100. In some embodiments, local management agent 332 may include a workspace engine suitable for instantiating and managing the operation of one or more workspaces 331A-N on IHS 100. As described, the capabilities of a workspace may be modified based on changes in the productivity and security contexts in which the workspace is operating.

Accordingly, the workload(s) in each of workspaces 331A-N may be hosted in a public cloud, a private cloud, a specific server, or locally hosted on IHS 100, depending on the context in which the workspace is operating. These allocations of workspace computing for each particular workspace 331A-N may be prescribed by the workspace definition that is used to build and operate each workspace.

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As described, the workspace definition may be created by workspace orchestration service 206 based upon context information provided by IHS 100, security targets for each workspace 331A-N, and productivity targets for each workspace 331A-N.

In some embodiments, local management agent 332 may be configured to host, launch, and/or execute a workspace hub 327 that provides a launch point 203 by which user's initiate workspaces through the selection of managed data and resources. In various embodiments, launch point 203 may be an agent, application, special-purpose workspace or web portal that provides an interface by which a user may select from an aggregated collection of data sources, applications, calendars, messages or other managed information or resources that are available to the user of IHS 100 via operation of a workspace as described herein. In various embodiments, the launch point 203 may be provided in the form of textual, graphical and/or audio user interfaces that allow a user of IHS 100 to select available data and/or resources. In some embodiments, workspace hub 327 may utilize a local environment management module 328 in providing the workspace interface that is presented to the user on IHS 100 and doing so in a consistent manner across workspaces 331A-N. Workspace hub 327 may also include a local intelligence logic or engine 329 used to support modeling the use of IHS 100 in order to improve characterization of the actual risk associated with a risk context. User authentication and access control operations may be performed by a local identify module 330 that may interface with trusted controller 341 in providing user authentication.

In some cases, each instantiated workspace 331A-N may be an environment that provides a user with access to requested data or applications, where the environment may be isolated in varying degrees from the hardware and software of IHS 100 based on the security context and productivity context in which each workspace 331A-N is operating. In some instances, the selection of a data source or resource that are available to user via launch point 203 may result in launching a new workspace. For instance, if a user launches a browser through selection of an icon displayed by launch point 203, a new workspace may be created and launched according to a workspace definition that has been selected for providing the user access to a web browser in the security and productivity contexts in which the request has been made. In a scenario where the user double clicks on a confidential presentation file available from a data source that is provided by launch point 203, an additional workspace may be instantiated with a presentation application providing access to the requested presentation file, where this new workspace is created based on a workspace definition that provided appropriate security for access to the confidential presentation). In other instances, a selection of the presentation file by a user may result in the presentation being made available through the existing workspace, in some cases using the existing workspace definition and, in other cases, using a workspace definition that has been modified to support the requested access to the confidential presentation file.

Although workspaces 331A-N supported by IHS 330B may each be isolated to varying degrees from the hardware and/or software of IHS 100 and from each other, a user of IHS 330B may expect to be able to operate the multiple workspaces 331A-N in a manner that allows content to be transferred between the different workspaces 331A-N. For instance, a user may select a portion of the data displayed in workspace 331A and utilize OS or other workspace functions to copy the data for copying to workspace 331B.

In various embodiments, local management agent **332** may operate in full or in part on secure platform **345** hosted by trusted controller **341** that operates independent from the OS of IHS **100**. In some embodiments, all or part of local management agent **332** may operate as trusted components of the OS of IHS **100**. To execute the various operations described herein, local management agent **332** may include command monitor **334** configured to provide instrumentation to receive commands from workspace orchestration service **206** and thus enable access to IHS **100**. Local management agent **332** may also include telemetry module **335** that may be configured for communicating collected information to workspace orchestration service **206**, including reporting changes in context that may warrant adjustments to workspaces **331A-N**. Data aggregator **336** may track all of the data source and other resources (e.g., applications, local or cloud-based services) that may be provided to the user via a workspace.

Local management agent **332** may utilize resource manager module **337** that is configured to manage access to data, network configuration, such as for VPNs and network access, identity information, access control, and resource provisioning services. Security module **338** may be configured to provide various security services. BIOS interface **339** may provide a secure BIOS interface used for accessing and managing credentials in secure object storage. BIOS analytics module **340** may be configured to perform forensic services for BIOS telemetry and health assessments. Persistence module **346** may be configured to support persistence of applications entitled at a POS or assigned by administrators and supported with required license tracking. Workspace attestation module **333** may provide a platform centric service layer on top of a container engine provided by local management agent **332** and may be used to measure and attest workspaces **331A-N** in any suitable manner defined or orchestrated by condition control **305**.

As part of secure platform **345**, native management module **347** may be configured to enable out-of-band management interface with workspace orchestration service **206**, where this OOB interface operates independent from the OS of IHS **100**. In some embodiments, the OOB management interface supported by native management module **347** may be utilized by the device configuration services **316** of the workspace orchestration service to access the secure platform services **345** of IHS **100**.

Digital device ID module **348** may provide a unique, un-spoofable, cryptographically bound identifier. In embodiments supporting a secure platform **345**, secure embedded controller **341** may be a hardened hardware module that may include a root of trust module **342** configured as a trusted data store and, in some cases for cryptographic processing, that may be trusted within a cryptographic system. A device attestation service **343** may be configured to perform device assurance and trust services (e.g., secure BIOS and secure boot, etc.). A secure object store **344** may be provided that is configured to lock and access keys, hashes, and/or other secrets in an EC and/or trusted platform module (TPM).

In some scenarios, IHS **100** may be provisioned by a manufacturer that also controls manufacturer integration components **317**, workspace attestation module **333** may operate in conjunction with secure object store **342**, authenticated BIOS module **339**, and/or digital device identity module **348**, etc., to further secure and/or control productivity features available in any of workspaces **331A-N** based upon hardware devices and settings unique to that IHS and/or designed specifically by that manufacturer.

To further illustrate how the systems and methods described herein operate to modernize workspace and hardware lifecycle management in an enterprise productivity ecosystem, three non-limiting use-cases or examples are discussed in turn below.

Use-Case A

In use-case A, a given user may request access to a protected data source on the enterprise's premise using a corporate-owned and imaged notebook, such configured as described with regard to IHS **100** of FIG. 1 and client IHS **100** of FIG. 3B.

In response to the request, a local management agent **332** operating on the user's notebook retrieves information describing the context and calculates security and productivity targets based on the determined context. In this use-case, the local management agent may have been installed by IT, and it may be running in the background as a service. The confidential data may be associated with the local management agent on the local machine, based on file classification (e.g., file metadata/type/properties/permissions, folder location, encrypted region, etc.). Moreover, the local management agent may continuously collect context information and send it to the orchestration service for use in scoring the risk and productivity of the workspace (this may also be done at the time of the user's access request or indication of intent).

When the user selects the confidential data, such as via a selection via the OS of the notebook, the local management agent notifies the workspace orchestration service of the request and for a workspace definition for a workspace by which the user may be provided access to the confidential data.

In this example, the workspace orchestration service may score an overall security risk to have a value of "2," using a weighed, machine learning, or artificial intelligence algorithm, based upon the following context information or inputs, each of which is also given as a risk metric based upon a selected policy: locale: 1 (safe locale); user persona: 1 (known high-confidence in a reasonably sophisticated user classification—a user whom historically does not click on phishing emails); network risk: 1 (low risk because of on premise, wired connection detected); device risk: 1 (high level of control because of corporate owned/managed platform, known versions, security features enabled, etc.); regulatory: 1 (based on user, data, location combinations—e.g., No restrictions with respect to General Data Protection Regulation or "GDPR," Health Insurance Portability and Accountability Act "HIPAA," Payment Card Industry "PCI," technology export, etc.); and data type: 8 (a confidential datafile is being requested).

The workspace orchestration service may also calculate a productivity score to have a value of "9," using a weighed, machine learning, or artificial intelligence algorithm, based upon the following context information or inputs, each of which is also given as a resource metric based upon a selected policy: locale: 10 (office); user persona: 9 (a "skilled" classification based upon advanced compute tasks, proficiency, and/or speed); network speed/latency: 10 (fast, wired, Gigabit Ethernet, or direct to internal network); device performance: 8 (fast, expensive CPU, memory, graphics, but storage only needs—e.g., <10 GB); and data type: 10 (the local, confidential file is easy to read/write with low latency and high performance on local storage).

Second, based upon the security score and/or context information, the workspace orchestration service builds a workspace definition file having any suitable structure with workspace definition attributes in a machine-readable for-

mat (e.g., JSON name-value, XML structured, etc.). In this example, the security target may be deemed to have a value of “1” based upon a combination of attributes values representing loads, needs, or demands on security controls and containment features that may include: threat monitoring: 1 (low demand); threat detection: 1 (low demand); threat analytics: 1 (low demand); threat response: 1 (low demand); storage confidentiality: 2 (low); storage integrity: 2 (low); network confidentiality: 1 (low); network integrity: 1 (low); memory confidentiality: 1 (low); memory integrity: 1 (low); display confidentiality: 1 (low); display integrity: 1 (low); user authentication: 1 (low, basic password is fine, non-multifactor authentication or “MFA,” no session expiration); IT administrator scope: 1 (administrator manages remotely but does not need heavy remediation software; and regulatory compliance: 1 (no GDPR, No HIPAA, no PCI, no tech export restriction, etc.).

Based upon the productivity target and/or context information, a productivity target for the workspace definition may be deemed to have a value of “9” (defining a high-quality, responsive user experience) based upon a combination of attribute values representing productivity requirements as follows: local storage: 7 (partial hard drive control, some storage reserved for IT load); CPU access: 10 (unlimited); local graphics: 10 (unlimited); and application stack: 10 (can use applications, install applications that the user needs, give them administrator rights, etc.).

Third, after the workspace definition is complete, the workspace orchestration service and the local management agent may assemble the workspace and instantiate it for the user. For example, the local management agent may receive definition files (e.g., JSON, XML, etc.) from the orchestration service, and it may parse the file to implement security risk controls such as: threat monitoring: 1 (local management agent does not install threat, detection, and response or “TDR” software); threat detection: 1 (local management agent does not install TDR software); threat analytics: 1 (orchestration does not need to gather detailed telemetry from the system, OS will not be enrolled in logging); threat response: 1 (local management agent does not install security threat response agent); storage confidentiality: 2 (local management agent deploys a local file-system encryption product that the user can optionally enable on specific files as needed with right-click context menus); storage integrity: 2; network confidentiality: 1 (local management agent confirms basic firewall configuration is correct—e.g., IT GPO-controlled); network integrity: 1; memory confidentiality: 1 (local management agent confirms configuration—e.g., No SGX, TXT, or container/sandbox software deployed); memory integrity: 1; display confidentiality: 1 (local management agent confirms graphics drivers installed, privacy screen and camera optionally managed by user); display integrity: 1; user authentication: 1 (local agent confirms basic GPO password rules are configured, and met by user—e.g., number of characters, no session expiration, etc.); IT administrator scope: 1 (local agent runs with system privilege, confirms IT admin accounts are listed in local admin user group—e.g., per GPO); and regulatory compliance: 1 (local agent does not install any compliance assistance software).

After confirming the configuration, the workspace orchestration service and the local management agent may give the user access to the requested local confidential file, and the user may begin working in a newly created workspace.

Use-Case B

In use-case B, a user may request access to a confidential datafile while at a coffee shop using an open public network

and an IT-managed/owned PC, such configured as described with regard to IHS 100 of FIG. 1 and client IHS 100 of FIG. 3B.

First, a local management agent (332) executed by client IHS 100 retrieves the requested context and calculates security and productivity scores based on context. In this use-case, the local management agent may have been installed by IT, and it may be running in the background as a service. The confidential data may be kept on a shared IT-managed network resource on-premises (e.g., back in a main corporate office), and the local management agent may be responsible for monitoring when this data path is requested by the user (e.g., the user hits a specific URL, IP, etc.). Moreover, the local management agent may continuously collect all context and send it to the workspace orchestration service to assist in scoring processes later (this may also be done at the time of the user’s access request or indication of intent, rather than a continuous collection).

When the user selects the desired confidential datafile, client IHS 100’s OS calls the local management agent associated with the path to the confidential datafile and calls back to a remote workspace orchestration service (206) to request a workspace definition.

In this example, the workspace orchestration service may score an overall security risk to have a value of “4,” using a weighed, machine learning, or artificial intelligence algorithm, based upon the following context information or inputs, each of which is also given as a risk metric based upon a selected policy: locale: 5 (public, safe country); user persona: 5 (new user, classification data does not exist yet); network risk: 5 (medium, public but common location, wireless connection detected); device risk: 1 (high level of control, corporate owned/managed platform, known versions, security features enabled, etc.); and regulatory: 1 (based on user, data, location combinations—e.g., no restrictions with respect to General Data Protection Regulation or “GDPR,” Health Insurance Portability and Accountability Act “HIPAA,” Payment Card Industry “PCI,” technology export, etc.).

The workspace orchestration service may also calculate a productivity score to have a value of “5,” using a weighed, machine learning, or artificial intelligence algorithm, based upon context information or inputs, each of which is also given as a resource metric based upon a selected policy. For instance, security contexts inputs may include: locale: 6 (remote location but in USA major city, in a public area, non-employees are within visual/audio range of device); user persona: 5 (unknown confidence “null” classification, uses default onboarding assumptions); network speed/latency: 4 (medium, wireless but AC on shared network); and device performance: 8 (fast, expensive CPU, memory, graphics, but storage only needs <10 GB).

Second, based upon the security score and/or context information, the workspace orchestration service builds a workspace definition file having any suitable structure with workspace definition attributes in a machine-readable format (e.g., JSON name-value, XML structured, etc.). In this example, a security target may be deemed to have a value of “4” based upon a combination of attributes values representing loads, needs, or demands on security controls and containment features as follows: threat monitoring: 4 (medium demand); threat detection: 4 (medium demand); threat analytics: 4 (medium demand); threat response: 4 (medium demand); storage confidentiality: 4 (medium); storage integrity: 9 (high); network confidentiality: 5 (medium); network integrity: 2 (low); memory confidentiality: 4 (medium); memory integrity: 8 (high); display confidentiality: 7 (me-

dium/high—worried about “shoulder surfers” reading data from an adjacent seat or table nearby, public location) display integrity: 2 (low); user authentication: 4 (medium, two-factor authentication using a hardware token, session expiration upon sleep, screen lock, or logout); IT administration scope: 3 (administrator can monitor, manage, and remediate remotely if the user calls them for help with IT issues); and regulatory compliance: 1 (no GDPR, No HIPAA, no PCI, no tech export restriction, etc.).

Based upon the productivity target and/or context information, a productivity target for the workspace definition may be deemed to have a value of “7” (defining a high-quality, responsive user experience) based upon a combination of attribute values representing productivity requirements as follows: local storage: 7 (partial hard drive control, some storage reserved for IT load); CPU access: 10 (unlimited); local graphics: 10 (unlimited); and application stack: 7 (can use applications, can install some IT-approved applications that the user needs, but no administrator rights, because the user cannot be trusted to install only valid/safe productivity software, but can install pre-approved IT applications as needed).

Third, after the workspace definition is complete, the workspace orchestration service and the local management agent may assemble the workspace and instantiate it for the user. For example, the local management agent may receive definition files (e.g., JSON, XML, etc.) from the orchestration service, and it may parse the file to implement security risk controls such as: threat monitoring: 5 (local management agent installs or confirms prior installation/configuration of TDR software); threat detection: 5 (local management agent installs or confirms prior installation/configuration of TDR software); threat analytics: 5 (orchestration confirms telemetry is accessible, OS will be enrolled in logging if not already enrolled); threat response: 2 (local management agent downloads but does not run remote incident response application—preparation in case incident is detected); storage confidentiality: 5 (local management agent deploys a local container technology, such as sandbox, with restricted “save” permissions such that the confidential files will not be allowed to save locally on the PC, but can be accessed as long as the session is active in memory); storage integrity: 5; network confidentiality: 5 (local management agent steps up firewall protections, disabling all unnecessary ports, and establishes a VPN back to the corporate office for protecting traffic to the local sandbox); network integrity: 5; memory confidentiality: 5 (local management agent configures sandbox container to isolate application and data from other applications/threats that may infiltrate the host OS); memory integrity: 5; display confidentiality: 7 (local management agent confirms graphics drivers installed, enforces privacy screen and uses camera to detect specific onlooker threats); display integrity: 7; user authentication: 4 (local agent confirms basic GPO password rules are configured, and met by user—e.g., number of characters, no session expiration, etc., but also adds in a requirement for hardware token to log in and again to establish network); IT administrator scope: 4 (local agent runs with administrator and remote access privilege, confirms IT admin accounts are listed in local admin user group—e.g., per GPO); and regulatory compliance: 4 (local agent installs state specific rule enforcement or monitoring software).

After confirming the configuration, the workspace orchestration service and the local management agent may give the user access to the requested local confidential file, and the user may begin working in a newly created workspace.

Use-Case C

In use-case C, a user may request access to a confidential datafile in a web hosted remote portal using a browser from Kazakhstan, while at an internet café with a borrowed/rented PC, such configured as described with regard to IHS 100 of FIG. 1 and client IHS 100 of FIG. 3B, on an open WiFi network.

First, a remote workspace orchestration service (332) intercepts the access request and evaluates the browser and user context, and calculates security and productivity scores. In this use-case, there is no local management agent; all that is known is the browser and any telemetry returned or garnered through the HTTP/S session. Assume, for sake of this example, that the confidential data may kept on a shared IT-managed network resource on-premises (e.g., back in a main corporate office) and that the datafile will remain there with only remote rendering/access privileges. Web-based context may be gathered through the browser session or supplied by the user. Moreover, user context may also be collected for the workspace orchestration service through alternate side-channels (e.g., travel calendar information, recent user billing activity on corporate credit card, phone call logs, and/or location data).

When the user selects the desired confidential datafile from the web browser, the back-end web server infrastructure calls back to the workspace orchestration service to request a workspace definition.

In this example, the workspace orchestration service may score an overall security risk to have a value of “9,” using a weighed, machine learning, or artificial intelligence algorithm, based upon the following context information or inputs, each of which is also scored as a risk metric based upon a selected policy: locale: 9 (Kazakhstan); user persona: 1 (user was expected to be there, the timing seems right based upon past logins, and he has a biometric watch communicator proving he is alive, himself, and located where he says he is—so that IT can always trust him); network risk: 9 (high, public and in a very obscure place); device risk: 9 (zero trust); and regulatory: 8 (based on user, data, location combinations).

The workspace orchestration service may also calculate a productivity score to have a value of “5,” using a weighed, machine learning, or artificial intelligence algorithm, based upon the following context information or inputs, each of which is also given as a resource metric based upon a selected policy: locale: 3 (internet café device without great performance); user persona: 9 (known high-confidence and “skilled” classification—advanced compute tasks, proficiency, and speed); network speed/latency: 3 (low quality—Wireless G from a long way away); and device performance: 3 (have to be able to tolerably browse web pages but based on what the service believes the capabilities will be, the service should build simple ones).

Second, based upon the security score and/or context information, the workspace orchestration service builds a workspace definition file having any suitable structure with workspace definition attributes in a machine-readable format (e.g., JSON name-value, XML structured, etc.). In this example, a security target may be deemed to have a value of “9” based upon a combination of attributes values representing loads, needs, or demands on security controls and containment features as follows: threat monitoring: 10 (high demand, to be handled on the server side); threat detection: 10 (high demand, to be handled on the server side); threat analytics: 10 (high demand, to be handled on the server side); threat response: 10 (high demand, to be handled on the server side); storage confidentiality: 10 (high demand, to be

handled on the server side); storage integrity: 8; network confidentiality: 10 (high demand, to be handled on the server side); network integrity: 9; memory confidentiality: 10 (high demand, to be handled on the server side); memory integrity: 9; display confidentiality: 10 (high, “shoulder surfers” may read datafile from an adjacent seat or table nearby in a public location); display integrity: 9; user authentication: 10 (high, three-factor authentication using login, hardware token, and biometric satellite watch—session expiration and refreshes every 30 seconds); IT administrator scope: 8 (administrator may monitor, manage, and remediate remotely if the user calls them for help or anything unexpected happens); and regulatory compliance: 10 (all network traffic is securely monitored as will the data presented).

Based upon the productivity target and/or context information, a productivity target for the workspace definition may be deemed to have a value of “3” (defining a usable secure user experience primarily built for consumption and not productivity) based upon a combination of attribute values representing productivity requirements as follows: local storage: 1 (cache only); CPU access: 3 (build for limited expectations); local graphics: 3 (build for limited expectations); and application stack: 1 (web browser experience on a kiosk mode device, limited data entry capability, limited read access to need-to-know only information through VDI rendered kiosk).

Third, after the workspace definition is complete, the workspace orchestration service and remote cloud web portal (e.g., session the user logged into through the browser) may assemble the workspace and instantiate it for the user in the browser. For example, the web portal may receive definition files (e.g., JSON, XML, etc.) from the orchestration service, and it may parse the file to implement security risk controls such as: threat monitoring: 9 (data center based management agent installs or confirms prior installation/configuration of TDR software); threat detection: 9 (data center based management agent installs or confirms prior installation/configuration of TDR software); threat analytics: 9 (orchestration confirms telemetry is accessible, server hosting web server may be enrolled in logging if not already enrolled—user behavioral telemetry from side channels may also be continuously monitored for suspicious/anomalous activity); threat response: 10 (data center-based management agent sets up watchdog timer to kill session automatically without periodic check-ins from orchestration, user telemetry, and web browser); storage confidentiality: 9 (data center-based management agent builds a progressive web application that may be used to display the data through a secure TLS link—the data will be rendered but only the as-needed portions of visualization presented to the user, and nothing can be saved); storage integrity: 10; network confidentiality: 9 (route traffic through best effort to secure locations—do not allow anything except bitmap renderings through the enforceable network); network integrity: 4; memory confidentiality: 9 (web page viewer only—no data leaves the data center, no confidential input is taken from the rented PC, no keyboard input is allowed, and all input may be captured from randomized virtual keyboard using mouse click coordinates); memory integrity: 8; display confidentiality: 8 (best effort to ensure confidentiality—prompt user at least—adjustable font sizes, but defaults to small fonts, obfuscated text, etc.); display integrity: 2; user authentication: 9 (local agent confirms basic password rules are configured, and met by user—e.g., number of characters, no session expiration, etc., but also adds in a requirement for hardware token and biometric, satellite watch to log in and again to establish network,

requiring frequent reconfirmation from user); IT administrator scope: 7 (data center-based remote environment); and regulatory compliance: 8 (local agent does not exist but data center-based agent monitors/blocks data not appropriate).

After confirming the configuration, the workspace orchestration service and the local management agent may give the user access to the requested rendered data, and the user may begin working in a newly created workspace.

FIG. 4 is a diagram of an example of heterogeneous workload environment 400 configured to perform the migration of workloads across cloud services based upon IHS performance. In environment 400, physical machine or endpoint 402 may implement IHS 100 and devices 401 may implement certain components of IHS 100 such as, for instance, user input devices 111, sensors 112, optical drive 114, I/O ports 116, etc.

Hypervisor 403 is a software program executed by physical machine or endpoint 402 that creates and runs software-based containers, such as software-based container 407, for example. Although hypervisor 403 is shown as a type-1, native, or bare-metal hypervisor (running directly on hardware 402 to manage host OS kernel 404), in other implementations hypervisor 404 may be a type-2 or hosted hypervisor (running on top of host OS kernel 404).

To produce and/or manage a first type of workload, hypervisor 403 supports host OS kernel 404, which in turn enables native application 406 to execute using native binary files and/or library files (bins/libs) 405. To concurrently produce and/or manage a second type of workload, host OS kernel 404 also supports the execution of software-based container 407, where container application 408 executes using container bin/libs 409. Software-based Container 407 may include, for example, a sandbox instance, Virtual Machine (VM), docker, snap, PWA, VDI, etc.

To concurrently produce and/or manage yet a third type of workload, hypervisor 403 also enables the execution of hardware-based container 410. Within hardware-based container 410, local management agent 332 instantiates workspace 331A based on a workspace definition such that application 411 executes using container bin/libs 412 within the constraints of workspace 331A. In some cases, hardware-based container 410 may include a uni-mini kernel engine or the like (e.g., Hyper-V, INTEL Clear Container, etc.). Other types of workloads may also be supported in environment 400.

In some implementations, software-based container 407 may be configured to execute applications or workloads that do not require a high level of security, for example, because they are trusted, such as container application 408. Conversely, workspace 331A may be configured to execute applications or workloads that do require a high level of security, for example, because they are untrusted, such as application 411, and/or because a current context (e.g., user is presently at untrusted geographical location, IHS 100 is subject to different network conditions, etc.) matches one or more rules outlined in a policy. Additionally, or alternatively, workspace 331A may be configured to execute applications or workloads that require an OS kernel different than host OS kernel 404.

In some embodiments, when applications are distributed and/or deployed from a trusted source, software-based container 407 may be used as they generally have less overhead and provide higher containerized application density. Conversely, when applications are distributed and/or deployed from an untrusted source, hardware-based (hypervisor iso-

lated) workspace **331A** may be used, despite presenting a higher overhead, to the extent it provides better isolation or security.

Software-based container **407** shares the kernel of host OS **404** and UEFI services, but access is restricted based on the container's user privileges. Hardware-based container **410** and/or workspace **311A** may have a separate instance of OS and UEFI services. In both cases, containers **407** and **410** serve to isolate applications **408** and **411** from host OS kernel **404** and other applications.

In various embodiments herein, each of a plurality of end users such as, for example, employees of an organization, may operate one or more client IHSs or endpoints, each client IHS having a different hardware configuration and/or resources. Moreover, the organization may provide the plurality of users with one or more time and/or volume subscriptions or licenses to one or more cloud computing services for additional compute infrastructure.

Examples of cloud computing services usable with systems and methods described herein may include, but are not limited to: Software as a Services (SaaS), Infrastructure as a Service (IaaS), and Platform as a Service (PaaS), among others.

With SaaS, users can access, share, store and secure information in "the cloud." Examples of SaaS include, but are not limited to file storage and/or backup, web-based email, project management tools, etc. IaaS provides the infrastructure (e.g., storage servers and networking hardware, but also load balancing, application firewalls, etc.) that many cloud service providers need to manage their own SaaS tools and can often serve as a complete data center, eliminating the need for resource-intensive, on-site installations. PaaS serves as a web-based environment (e.g., database, operating system, programming language, etc.) where developers can build cloud applications.

In most cases, a cloud service provider (e.g., AMAZON WEB SERVICES, GOOGLE CLOUD, MICROSOFT AZURE, etc.) may offer different types of cloud services to match different types of workloads. Examples of types of cloud services include, but are not limited to: Artificial Intelligence (AI) or Machine Learning (ML), analytics, blockchain, compute, databases, developer tools, integration, Internet-of-Things (IoT), mixed reality, networking, security, storage, web, etc. Additionally, or alternatively, different cloud service types may be categorized as: general purpose, compute optimized, network optimized, memory optimized, etc. Each different type of cloud service may offer a different level of performance when executing a given workload.

Moreover, within each different type of cloud service, a cloud service provider may also offer different performance tiers. For example, a high-performance tier (e.g., tier 1) cloud service may allocate a first amount of cloud compute power to execute a workload, typically at a first cost. A mid-performance tier (e.g., tier 2) cloud service may allocate a second amount of cloud compute power smaller than the first amount to execute the workload, typically at a second cost smaller than the first cost. A low-performance tier (e.g., tier 3) cloud service may allocate a third amount of cloud compute power smaller than the second amount to execute the workload, typically at a third cost smaller than the second cost.

As part of a modern client experience, a user employs local and cloud resources to perform his day-to-day tasks to run workloads and applications on either cloud or local machine, for example, depending on context. Because clients predominantly use cloud resources on an hourly basis,

the ITDM has to be aware of the resources available from various cloud infrastructures in order to optimize the cost-per-user. In some cases, the same workload may be configured to run on cloud service instances of different types, and it may be executed at any available cloud service tier, at a different cost.

As the inventors hereof have recognized, however, the performance of a cloud service should be directly correlated with the performance of an IHS endpoint. When a workload is running on a high-performance cloud service instance, which can deliver the possible best user experience, the endpoint must be capable of taking advantage of the cloud service without sacrificing its own performance. Using the systems and methods described herein, should the endpoint's performance be negatively impacted by the high-performance cloud service and/or should the high-performance cloud service be in excess of than what the endpoint can effectively absorb, the workload may be migrated to a different lower performance provider, service, and/or tier instance, for a better overall end user experience and/or to preserve IHS resources (e.g., compute, memory, battery, etc.). Conversely, when a workload is running on a low-tier cloud service and an endpoint is capable of handling more bandwidth and/or has available compute capabilities, systems and methods described herein may migrate the workload to a higher performance provider, service, and/or tier instance.

For example, assume a scenario where a user is running a web-based MICROSOFT OFFICE **365** application and the ITDM originally provisioned the user to run that workload on a specific type of processor in hyper-threaded configuration, providing a better value proposition for most general-purpose workloads. If the user the user is running the application on a work IHS at a work location, with wired network connectivity, and AC power, for example, systems and methods described herein may migrate the workload to next higher tier cloud instance to deliver better end-user experience. However, if the user is running the application over a public WiFi access point, on an IHS with a display's brightness setting dimmed (to protect the battery from draining), for example, systems and methods described herein may migrate the workload to the lowest tier cloud instance to pair optimal performance with associated cost savings.

Still referring to FIG. 4, heterogeneous workload environment **400** may be configured to grade or rate workloads and to trigger the migration of cloud instances executing those workloads based upon contextual information indicative of an IHS's performance and/or capabilities. Particularly, ITDM agent **413** may be configured to enforce one or more migration policies (e.g., with migration rules and conditions rules encoded in XML files, etc.) retrieved from policy database **416**, and to trigger orchestration service **206** to perform a workspace and/or workload migration operation.

ITDM agent **413** may also be configured to request and/or receive telemetry data (e.g., related to a user, an application, hardware, etc.) from one or more of: software-based container **407**, hardware-based container **410**, Energy Estimation Engine (E3) or Advanced Configuration and Power Interface (ACPI) module **414**, or kernel service **415**.

Particularly, upon instantiation, kernel service **415** and/or ITDM agent **413** may collect telemetry data from the containers/workloads including, but not limited to: workspace telemetry tracked when a user is installing the workload, new driver installations, registry entries, user settings, hardware resources that a workload is accessing (e.g., stor-

age, flash memory, keyboard, mouse, etc.), location or geographic information, version information, battery/power consumption data, etc. Kernel service **415** and/or ITDM agent **413** may also receive telemetry data from E3/ACPI module **414** indicative of the IHS's capabilities and/or performance including, but not limited to, energy usage data per workspace/workload, global states, device states, processor states, performance states, etc.

When a user launches a workload, a new set of telemetry data may be tracked by kernel service **415** and/or ITDM agent **413**, including, but not limited to: the network activity of a container checking the back-end for subscription/version information, etc., a network access Uniform Resource Locator (URL), hardware resources that the workload is accessing (e.g., camera, storage devices, external peripheral devices, etc.), default settings when the user launches the workload, a temporary storage location where the user stores files, documents, or artifacts (e.g., .doc, .pdf, etc.) resulting from the execution of the workload, read/write activity on a storage device, a type of workload, the type of container executing the workload (e.g., a software-based container or a hardware-based container), I/O events, CPU/GPU performance statistics, etc.

ITDM agent **413** may also be configured to communicate with orchestration service **418** (e.g., a component of application services **301**) over network **417**. When orchestration service **418** receives a trigger from ITDM agent **413**, for example, in response to enforcement of a migration policy, orchestration service **418** may migrate an identified workload to a different performance provider, service, and/or tier instance. To that end, orchestration service **418** may be configured to communicate with one or more cloud service providers using their respective Application Programming Interfaces (APIs).

In some cases, orchestration service **418** may also be configured to use cloud provider APIs to obtain performance metrics on a cloud service instance, to estimate the next lower/higher performance level or grade of a type of cloud service instance, to determine whether to migrate a workload to a different provider, service, and/or tier based upon a migration policy identified in a workspace definition, to send control information to ITDM agent **413**, to serve as backend orchestration control point for cloud-native workload registration to cloud APIs (e.g., ZOOM-managed cloud APIs), and/or to communicate to the ITDM the next cloud instance the workload switched to (e.g., for billing purposes).

Orchestration service **418** may be further configured to provide a trigger to ITDM agent **413** to swap a particular workspace between different instance of the cloud or to perform other responsive actions (e.g., install another application, change a setting of an application in anticipation of workload migration, etc.) and/or to manage user authorization and authentication of cloud native applications during the migration to maintain the user session across environments.

Orchestration service **418** and/or ITDM agent **413** may be configured to generate a recommendation of which cloud provider, service, and/or performance tier to migrate a workload to based, at least in part, upon any of the telemetry data or context information discussed above, alone or in any combination. The telemetry data may be used to ascertain metrics indicative of a capability and level of utilization of a given IHS and/or IHS resource/component. Moreover, the difference between capability and utilization metrics may be indicative of a performance level of the IHS during execution of a workload.

In some cases, in response to the difference being greater than a threshold, orchestration service **418** may migrate the workload to another cloud-based service with higher performance than the cloud service. Conversely, in response to the difference being smaller than a second threshold (itself smaller than the first threshold), orchestration service **418** may migrate the workload to another cloud-based service with lower performance than the cloud service. As such, systems and methods described herein may migrate workloads to match the performance levels of a cloud service and an endpoint IHS.

FIG. 5 is a flowchart of an example of method **500** for migration of workloads across cloud services based upon IHS performance. In various embodiments, method **500** may be performed, at least in part, by orchestration service **418**, ITDM agent **413**, and kernel service **415** of heterogeneous workload environment **400** in FIG. 4.

At **501**, orchestration service **418** pushes a migration or configuration policy to ITDM agent **413**. At **502**, ITDM agent **413** pushes a subset of the configuration policy to kernel service **415**, for example, using the Kernel-Mode Driver Framework (KMDF).

In steady state **503**, at **504**, kernel service **415** collects telemetry data outlined in the subset of the configuration policy received from ITDM agent **413** (e.g., user, geo, OS registry, battery, etc.). At **505**, kernel service **415** sends aggregated telemetry data to ITDM agent **413**. At **506**, ITDM agent **413** further aggregates the telemetry data from kernel service with other telemetry data (e.g., from containers **407/410**, hypervisor **403**, E3/ACPI module **414**, etc.) and sends it to orchestration service **418**.

Examples of E3 data include, but are not limited to, for any workload: a process runtime, foreground or background state, AC/DC power consumption, I/O activity, utilization of CPU, GPU, memory, disk, network, etc. For instance, energy data from E3 may yield charts and/or tables relating, for each process or application workload (e.g., outlook.exe, powerpnt.exe, chrome.exe, etc.), data about energy consumption, screen on time, battery consumption, interactivity state, etc. of that workload, as measured by the OS. Additionally, or alternatively, for certain workloads, application-specific telemetry may be used (e.g., ZOOM telemetry, MICROSOFT Graph API, etc.).

At **507**, orchestration service **418** applies a configuration or migration policy to the telemetry data and at **508** it migrates a cloud-native workload to a higher/lower performance service as prescribed in the configuration or migration policy.

In some cases, at **507** orchestration service **418** may identify overutilized or underutilized IHSs (e.g., when a difference between the IHS's capabilities and utilization reaches threshold values defined in the migration policy) and migrate cloud-based workloads to a different provider, service, and/or performance tier. The migration policy may be identified or included in a workspace definition of the workspace handling a cloud-based workload, and it may be selected based upon any of combination of one or more of the aforementioned context information.

At **509**, orchestration service **418** notifies ITDM agent **413** of the migration and provides a trigger for other responsive actions (e.g., install another application, etc.). At **510**, orchestration service **418** sends another notification of the migration to an ITDM console (e.g., for cost monitoring), and at **511** ITDM agent **413** performs the other responsive actions, if any.

In some cases, to perform a workspace migration, cloud performance tiers may be updated up or down. To select the

proper tier based on the telemetry data, orchestration service 418 may employ machine learning (ML) algorithms for solving knapsack-type problems using various techniques, including dynamic programming, brute-force evaluations, etc. Examples of suitable ML algorithms may include, but are not limited to: linear regression, logistic regression, decision tree, naive Bayes, k-nearest neighbors (KNN), etc.

Accordingly, embodiments of systems and methods described herein may suggest recommendations for optimal balanced performance of endpoints and cloud services executing workloads, so that an ITDM may provide a modern client experience to users of an organization. In various embodiments, systems and methods described herein may provide the ability to grade workload orchestration migration between various cloud instances for optimal end-point performance.

It should be understood that various operations described herein may be implemented in software executed by processing circuitry, hardware, or a combination thereof. The order in which each operation of a given method is performed may be changed, and various operations may be added, reordered, combined, omitted, modified, etc. It is intended that the invention(s) described herein embrace all such modifications and changes and, accordingly, the above description should be regarded in an illustrative rather than a restrictive sense.

The terms “tangible” and “non-transitory,” as used herein, are intended to describe a computer-readable storage medium (or “memory”) excluding propagating electromagnetic signals; but are not intended to otherwise limit the type of physical computer-readable storage device that is encompassed by the phrase computer-readable medium or memory. For instance, the terms “non-transitory computer readable medium” or “tangible memory” are intended to encompass types of storage devices that do not necessarily store information permanently, including, for example, RAM. Program instructions and data stored on a tangible computer-accessible storage medium in non-transitory form may afterwards be transmitted by transmission media or signals such as electrical, electromagnetic, or digital signals, which may be conveyed via a communication medium such as a network and/or a wireless link.

Unless stated otherwise, terms such as “first” and “second” are used to arbitrarily distinguish between the elements such terms describe. Thus, these terms are not necessarily intended to indicate temporal or other prioritization of such elements. The terms “coupled” or “operably coupled” are defined as connected, although not necessarily directly, and not necessarily mechanically. The terms “a” and “an” are defined as one or more unless stated otherwise. The terms “comprise” (and any form of comprise, such as “comprises” and “comprising”), “have” (and any form of have, such as “has” and “having”), “include” (and any form of include, such as “includes” and “including”) and “contain” (and any form of contain, such as “contains” and “containing”) are open-ended linking verbs. As a result, a system, device, or apparatus that “comprises,” “has,” “includes” or “contains” one or more elements possesses those one or more elements but is not limited to possessing only those one or more elements. Similarly, a method or process that “comprises,” “has,” “includes” or “contains” one or more operations possesses those one or more operations but is not limited to possessing only those one or more operations.

Although the invention(s) is/are described herein with reference to specific embodiments, various modifications and changes can be made without departing from the scope of the present invention(s), as set forth in the claims below.

Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of the present invention(s). Any benefits, advantages, or solutions to problems that are described herein with regard to specific embodiments are not intended to be construed as a critical, required, or essential feature or element of any or all the claims.

The invention claimed is:

1. An Information Handling System (IHS), the IHS comprising:

a processor, and

a memory coupled to the processor, the memory having program instructions stored thereon that, upon execution, cause the IHS to perform operations comprising: determine a learned capability level of the IHS based on prior telemetry data received when a workload of a given type was executed by the IHS, wherein the prior telemetry data comprises historical Central Processing Unit (CPU) utilization, latency, and energy efficiency data of the IHS;

receive telemetry data indicative of a level of utilization of the IHS when an IHS workload of the client given type is executed by a first cloud service, wherein the telemetry data comprises CPU utilization, latency, and energy efficiency of the IHS;

compare the level of utilization with the learned capability level; and

determine whether migration of the workload should be activated based at least in part on a policy framework that accounts for latency and energy efficiency, further comprising:

in response to a determination that the level of utilization is greater than the learned capability level by at least a first threshold margin, activate migration of the workload to a second cloud service characterized by higher performance capacity and lower latency than the first cloud service; and

in response to a determination that the level of utilization is less than the learned capability level by at least a second threshold margin, activate migration of the workload to a third cloud service characterized by lower power consumption and lower thermal load than the first cloud service.

2. The IHS of claim 1, wherein the prior telemetry data or the telemetry data further comprises information indicative of at least one of: a software version, a driver installation, a registry entry, or a user setting.

3. The IHS of claim 1, wherein the prior telemetry data or the telemetry data further comprises information indicative of usage of a hardware resource accessed by the workload.

4. The IHS of claim 3, wherein the hardware resource further comprises a battery.

5. The IHS of claim 3, wherein the hardware resource further comprises at least one of: a storage, a memory, a keyboard, a mouse, a camera, or a display.

6. The IHS of claim 5, wherein the prior telemetry data or the telemetry data further comprises information indicative of a display setting.

7. The IHS of claim 1, wherein the prior telemetry data or the telemetry data further comprises information indicative of a temporary storage location where a user stores a file used by the workload.

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8. The IHS of claim 1, wherein the prior telemetry data or the telemetry data further comprises information indicative of at least one of: user proximity, a client IHS location, or a client IHS posture.

9. The IHS of claim 1, wherein the prior telemetry data or the telemetry data further comprises information indicative of at least one of: a network connection or a network activity.

10. The IHS of claim 1, wherein the prior telemetry data or the telemetry data further comprises information indicative of one or more workspace settings applied upon instantiation of a workspace configured to execute the workload.

11. The IHS of claim 10, wherein the one or more workspace settings further comprise an indication of whether the workspace is software-based or hardware-based.

12. The IHS of claim 1, wherein the prior telemetry data or the telemetry data further comprises information indicative of an IHS memory utilization.

13. The IHS of claim 1, wherein further program instructions, upon execution, cause the IHS to perform further operations comprising receive the prior telemetry data or the telemetry data from at least one of: an Energy Estimation Engine (E3) or an Advanced Configuration and Power Interface (ACPI) module of the IHS.

14. The IHS of claim 1, wherein further program instructions, upon execution, cause the IHS to perform further operations comprising execute a machine learning model configured and trained to determine the learned capability level based on telemetry data input to the machine learning model, wherein the telemetry data input comprises one or more of: read/write activity on a storage device, a type of workload, the type of container executing the workload, I/O events, CPU/GPU performance statistics, energy consumption data, screen on time data, battery consumption data, or workload interactivity state.

15. The IHS of claim 1, wherein at least one of the first threshold margin or the second threshold margin is selected by the policy framework.

16. The IHS of claim 15, wherein further program instructions, upon execution, cause the IHS to perform further operations comprising identify the policy framework based, at least, in part, upon a workspace definition associated with a workspace configured to execute the workload.

17. The IHS of claim 1, wherein the first cloud service, the second cloud service, and the third cloud service are provided by a single cloud service provider.

18. The IHS of claim 1, wherein at least two of the first cloud service, the second cloud service, and the third cloud service are provided by different cloud service providers.

19. A memory storage device having program instructions stored thereon that, upon execution by an Information Handling System (IHS), cause the IHS to perform operations comprising:

receive telemetry data indicative of a level of utilization of the IHS when a workload of the IHS is executed by a cloud resource belonging to a given performance tier,

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wherein the telemetry data comprises IHS Central Processing Unit (CPU) utilization data and IHS memory utilization data;

determine a level of performance of the IHS based at least in part on a machine learning model executed to determine the level of performance of the IHS, wherein the machine learning model is configured and trained to determine the level of performance of the IHS based at least in part on a difference between IHS capability and IHS utilization, based at least in part on the telemetry data;

determine a selected cloud performance tier that has a cloud performance level that corresponds to the level of performance of the IHS;

in response to a determination that the level of performance or utilization of the client IHS is greater by at least a first threshold margin than the cloud performance level of the selected cloud performance tier, migrate the workload to another cloud resource that has a cloud performance level less than the cloud performance level of the selected cloud performance tier; and in response to a determination the level of performance of the IHS is less by at least a second threshold margin than the cloud performance level of the selected cloud performance tier, migrate the workload to another cloud resource that has a cloud performance level greater than the cloud performance level of the selected cloud performance tier.

20. A method, comprising:

executing a workload by an Information Handling System (IHS) operating as a client IHS, using a first cloud service having a cloud performance level corresponding to a given cloud performance tier;

receiving, by a cloud service orchestrator IHS from the client IHS, telemetry data comprising client IHS Central Processing Unit (CPU) utilization data, client IHS latency data, and client IHS energy efficiency data measured at least in part by an Operating System (OS) kernel of the client IHS while the client IHS executes the workload using the first cloud service;

determining, by the cloud service orchestrator IHS, a level of performance of the client IHS based at least in part on executing a machine learning model configured and trained to determine the level of performance of the client IHS based at least in part on a difference between client IHS capability and client IHS utilization, based at least in part on the telemetry data; and

in response to determining, by the cloud service orchestrator IHS based at least in part on the telemetry data and the machine learning model, a change in the level of performance of the client IHS, migrating the workload being executed based at least in part on transferring a containerized instance of the workload to a second cloud service having a cloud performance level corresponding to a cloud performance tier different from the given cloud performance tier.

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